Keep, Measure, Erase

What a Quantum Loop Does with Its History

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1 Introduction

Suppose we want to know how many iterations a loop went through before halting, and we want that number without changing its output or what the loop actually did. In a classical program we can just allocate and increment a private counter as the body goes through each iteration and keep or ignore it when the loop finishes. This remains a valid way to get the iteration count even if the body tosses a coin to decide whether to continue. A terminating execution still goes through the body a definite number of times, so recording that number changes neither its returned value nor the probability of obtaining that value.

For a quantum program, recording the number of iterations can change the loop's output behaviour. A body operating on a quantum guard may send amplitude both out of the loop and back for another call. If the contributions leaving after different numbers of calls reach the same output, their complex amplitudes add before a measurement assigns concrete probabilities. A counter puts these contributions alongside orthogonal labels, which distinguish them even if nobody reads the counter. Ignoring it means observing only the returned data, whose state is obtained by tracing out that register. This removes the terms between different count labels. We ask when recording and discarding the count preserves every observation on the returned data: when does the loop with a record give the same data as coherent feedback?

We start with a finite unitary body $U : X \oplus Y \to X' \oplus Y$, connecting its output Y to its input Y. The established direct-sum trace adds what leaves after every possible number of returns, as in reversible iteration and geometry of interaction [1, 12, 13]. On finite unitary bodies it gives a unitary external map even when infinitely many rounds contribute [3, 4]. We use this trace to interpret coherent feedback (Section 3).

We use symmetric pattern matching to write programs over the finite sum and product types of Π (Section 2). Classical Π expresses finite bijections; its quantum extensions express unitary gates, with approximate universality for UΠ [11, §§ 3.1–3.2]. Our finite clause presentation admits exact complex coefficients, and we add trace for coherent feedback. Following *Quantum Information Effects* [11, §§ 3–4], we add allocation to supply a counter for the recording loop, and hiding to discard outputs and derive the guard measurement used on each turn [21, § 3] (Section 4). Allocation lets a program return a counter with its data as a norm-preserving isometry. Hiding that register gives a channel on the remaining data. A finite cyclic counter records the count modulo its size, and an exact count is their countable semantic limit. Discarding the exact record gives the channel of measuring the guard on every call.

We write L for the coherent loop and x_k for the linear map that takes an input u to the amplitude contribution $x_k u$ leaving after k returns. In Section 5 we give a finite test for equality of the measured channel and $\rho \mapsto L\rho L^\dagger$, permitting replacement in any channel context, including data entangled with a reference the caller keeps. It passes exactly when every exit is a scalar multiple of L, and checking the first $m+1$ exits suffices, with $m = \dim Y$. Then the record has a fixed state independent of the data. Kraus proportionality for finite families is known [20, Cor. 2.23]; our comparison connects it to an exit sequence, a finite check and a typed replacement.

This leaves a second question: what can we learn about the rounds of a loop that keeps no count (Section 6)? A phase on the return wire marks each passage, giving exit k the factor $e^{ik\varphi}$. The phase-dependent gate therefore has the original exits x_k as its Fourier coefficients. Averaging its

channels recovers the measured loop, and averaging its relative phase derivative gives the measured mean count. We use transfer functions of unitary systems from operator theory [5] and identities from monitored recurrence, which repeatedly evolves and tests for return to a chosen subspace [6]. These averages describe the measured program; at a fixed phase, coherent feedback still adds exits from different rounds. We follow one rotation from its clauses through feedback, agreement and the phase calculation.

2 Finite programs in iso

An iso $\omega : a \leftrightarrow b$ is a reversible program from data type a to data type b. We use the finite types

0	1	$a + b$	$a \otimes b$
no values	one value $*$	a tagged value from a or b	a pair from a and b.

These are the sum and product types of Π, including the empty and unit types [11, § 3.1]. Write $\mathscr{B}(a)$ for the basis values of a and $|s\rangle$ for the basis vector of $s \in \mathscr{B}(a)$. For example, $\mathscr{B}(1) = \{*\}$ and $1 + 1$ has values left $*$ and right $*$.

2.1 Syntax

A clause $v \leftrightarrow e$ matches an input pattern and binds its variables in e. Patterns v use unit $*$, variables x, tags left v, right v and pairs $\langle v_1, v_2 \rangle$; p, q use unit, variables and pairs. For finite index sets and $\alpha_i \in \mathbb{C}$, expressions form superpositions and apply local isos:

$$\begin{aligned} e &::= v \mid \textstyle\sum_i \alpha_i e_i \mid \text{let } p = \omega\, q \text{ in } e, \\ \mathscr{T} &::= \{v_i \leftrightarrow e_i\}_{i \in I}, \\ \omega &::= f \mid \mathscr{T} \mid \omega; \omega \mid \omega \oplus \omega \mid \omega \otimes \omega \mid \omega^\dagger. \end{aligned}$$

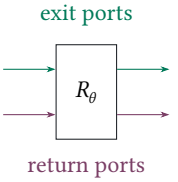
A name f refers to an earlier checked definition. Sequence ; runs left first; $\oplus$ acts on alternatives, $\otimes$ on simultaneous factors, and dagger reverses the iso. In a local application, q assembles an argument and p binds the result for e. A variable may bind any type, while unit and pair patterns make this result binding exhaustive.

The executable names `Unit` and `unit` mean type 1 and value $*$. The type `Qubit` is $1 + 1$, with `0` = left $*$ and `1` = right $*$. We use `exit` and `back` for the outer `left` and `right` tags. At type $a + c$, they carry a and c values; the tag alone neither stops nor repeats a computation. Thus `exit 0` expands to left (left $*$). Brackets group a superposition and a dot scales the expression that follows it. Closed application is written $\omega\, t$, where t is a closed expression or another closed application. The executable spelling is `omega on t`; `on` binds less tightly than sequence, so `f; g on t` applies the whole sequence to t.

The abbreviation `round` $f : a + c = \mathscr{T}$ declares an open body $f : (a + c) \leftrightarrow (a + c)$; it specializes the external input and output to the same type.

For our running example, suppose we want each call to rotate a guard qubit through $\theta = \arccos(3/5)$. Choose 0 as the exit tag and 1 as the return tag, with a unit payload on each. The rotation has $\cos\theta = 3/5$ and $\sin\theta = 4/5$, giving

```
round guardRound : Unit + Unit = {
  exit unit ↔ [ 3/5 . exit unit + 4/5 . back unit];
  back unit ↔ [-4/5 . exit unit + 3/5 . back unit]
}
```



Its type is $(1 + 1) \leftrightarrow (1 + 1)$. On input `exit unit`, one call returns $(3/5)|\text{exit } *\rangle + (4/5)|\text{back } *\rangle$. The closing brace ends this definition; it does not feed anything back. Both payloads are unit, so

this example needs no data qubit beside its guard. Section 3 will connect the return output to the return input.

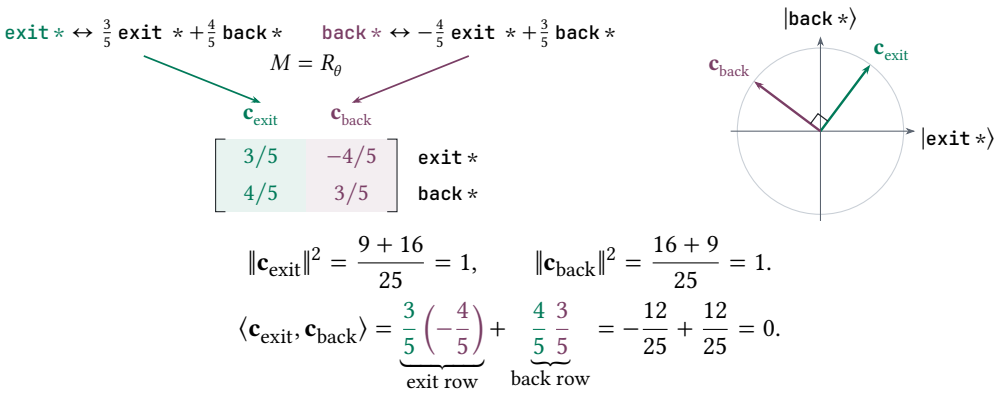
To evaluate a whole iso, we first replace each checked clause table $\mathcal{T}$ by an atom $[M]$ containing its matrix. This gives a smaller language of *matrix combinators*:

$$\kappa ::= [M] \mid \kappa;\kappa \mid \kappa \oplus \kappa \mid \kappa \otimes \kappa \mid \kappa^{\dagger}.$$

Thus ω is a source program, κ is a program built from matrix atoms, and M is the matrix computed by that program.

2.2 Statics

Match each basis input s against its unique clause and evaluate the output expression under those bindings. Its vector becomes column s of M: entry M_{ts} is the amplitude from input s to output t. Admission requires linearly typed expressions, a partition of the input basis, equal input and output basis sizes, and $M^{\dagger}M = I$. Dagger is conjugate transpose, so the diagonal entries test squared column norms and the off-diagonal entries test inner products. Together these make the columns an orthonormal basis. Figure 1 performs this check on the two clauses of `guardRound`.



$$\|\mathbf{c}_{\text{exit}}\|^2 = \frac{9+16}{25} = 1, \qquad \|\mathbf{c}_{\text{back}}\|^2 = \frac{16+9}{25} = 1.$$

$$\langle \mathbf{c}_{\text{exit}}, \mathbf{c}_{\text{back}} \rangle = \underbrace{\frac{3}{5}\left(-\frac{4}{5}\right)}_{\text{exit row}} + \underbrace{\frac{4}{5}\frac{3}{5}}_{\text{back row}} = -\frac{12}{25} + \frac{12}{25} = 0.$$

Fig. 1. Clauses become colored columns. Their real output vectors have unit length and zero inner product; the products compare the same output row. Complex columns use conjugation of the first factor.

The judgment $\Sigma; \Delta \vdash e : b$ uses earlier iso definitions in Σ and variables in the linear context Δ. Each sum alternative uses the same variables once; pairs and local applications split disjoint contexts, and result patterns bind fresh variables. A checked iso has judgment $\Sigma \vdash \omega : a \leftrightarrow b$. Matrix atoms use the same column test; the combinators match their boundary types. A declaration cannot call itself. It enters Σ only after checking. Admission and evaluation are defined together, recursing only into earlier definitions or proper subprograms.

The clauses and linear contexts adapt symmetric pattern matching [19, §§ 3–5] [15, Ch. 3]. Appendix A gives the inherited rules in full.

2.3 Dynamics

Write $\gamma \vdash e \Downarrow \psi$ for evaluation under a basis assignment. A value gives $|v[\gamma]\rangle$ and a sum adds its weighted outputs. A local application passes every weighted result to its continuation and adds the resulting vectors, without measurement or normalization.

Elaboration $\mathscr{E}_\Sigma(\omega) = \kappa$ replaces a checked table by its atom and unfolds a name in its earlier environment. It commutes with sequence, sum, tensor and dagger. For $\kappa \Downarrow M$ and $\lambda \Downarrow N$,

$$[M] \Downarrow M, \qquad \kappa; \lambda \Downarrow NM.$$

Sum, tensor and dagger use the corresponding matrix operations. Write $\Sigma \vdash \omega \Downarrow M$ when its elaboration evaluates to M. A closed term applies this matrix to its evaluated input vector: if $t \Downarrow \psi$, then $\omega\, t \Downarrow M\psi$.

2.4 Denotation

A type denotes $[\![a]\!] = \mathbb{C}^{\mathscr{B}(a)}$, with its basis values orthonormal. Thus 0 and 1 denote the zero space and $\mathbb{C}$, and sum and product denote Hilbert sum and tensor. A checked iso denotes the linear extension of its evaluated columns. The column check makes this map unitary; Appendix A proves the interpretation and the inherited combinator cases.

3 From a round to its exits

We now add the operation that lets us connect a body's returning ports and follow an input through the resulting feedback.

Syntax. Extend the source and combinator grammars with $\mathrm{trace}_c\, \omega$ and $\mathrm{trace}_c\, \kappa$, respectively. Here ω is a source iso and κ is a matrix combinator program. In either syntax, the constructor connects the body's output summand of type c to the input summand of the same type. QWhile uses this typed feedback interface for its do-while combinator [3, § 5.1].

Statics. The external input and output remain after this connection:

$$\frac{\Sigma \vdash \beta : a + c \leftrightarrow b + c}{\Sigma \vdash \mathrm{trace}_c\, \beta : a \leftrightarrow b} \qquad \frac{\kappa : a + c \leftrightarrow b + c}{\mathrm{trace}_c\, \kappa : a \leftrightarrow b}.$$

These are typing rules: they specify the remaining input and output types, without computing the loop. The subscript c is the return *type*, not a value tag. For a named `round`, the executable spelling `trace f` infers this type from its declaration. Elaboration preserves the connection:

$$\mathscr{E}_\Sigma(\mathrm{trace}_c\, \omega) = \mathrm{trace}_c\, \mathscr{E}_\Sigma(\omega).$$

Dynamics. Let $\beta : a + c \leftrightarrow b + c$ be the source body. Elaboration gives the matrix program $\kappa = \mathscr{E}_\Sigma(\beta)$, and evaluating that program, $\kappa \Downarrow U$, gives its matrix. For example, `guardRound` elaborates to $[R_\theta]$, which evaluates to R_θ. The symbol κ below always denotes a program.

Put $X = [\![a]\!]$, $X' = [\![b]\!]$ and $Y = [\![c]\!]$. To divide U into four blocks, group its input columns and output rows by their outer tags. Evaluating a source input `exit` s supplies the A and C entries of its column; evaluating `back` q supplies the B and D entries:

	input exit s	input back q
output exit t	A_{ts}	B_{tq}
output back r	C_{rs}	D_{rq}

$s \in \mathscr{B}(a), \quad t \in \mathscr{B}(b), \quad q, r \in \mathscr{B}(c).$

Thus $A : X \to X'$ is the immediate external action, $C : X \to Y$ sends an external input back, $B : Y \to X'$ lets a returning input leave, and $D : Y \to Y$ sends it back again. On a general vector, these same entries give

$$U\begin{pmatrix} u \\ y \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}\begin{pmatrix} u \\ y \end{pmatrix} = \begin{pmatrix} Au + By \\ Cu + Dy \end{pmatrix}. \tag{3.1}$$

Here $u \in X$ comes from outside and $y \in Y$ enters on the return port. The rows give external output $Au + By \in X'$ and returning output $Cu + Dy \in Y$. The body is *open* because its returning input

and output are still separate ports. Closing them gives the loop. Unitarity and the common finite Y imply $\dim X = \dim X'$. For `guardRound`, the first clause gives $A = 3/5$ and $C = 4/5$; the second gives $B = -4/5$ and $D = 3/5$.

The first call on external input u receives $(u, 0)$: here 0 is absent amplitude on Y, not a prepared basis state $|0\rangle$. Only Cu supplies the next call, with zero external input. Figure 2 follows three calls. These are components of the original amplitude, without normalization; the block inclusions are ambient linear maps, not reversible gates.

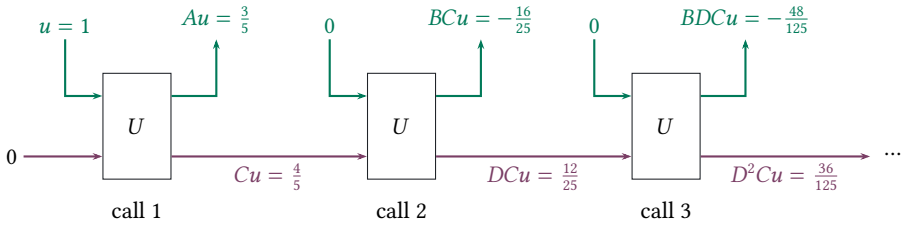


Fig. 2. Three calls, with scalar values for $U = R_\theta$ and $u = 1$. Purple return components alone supply the next call; later external inputs have zero amplitude.

Define $x_k : X \to X'$ to be the exit map after k returns. Thus count k means $k + 1$ body calls, including the first; adding one to the label gives the number of iterations from the introduction. The immediate exit is A. To exit after $k + 1$ returns, a contribution first enters Y through C, makes k further passages through D, and finally leaves through B. Its exit maps are therefore

$$x_0 = A, \qquad x_{k+1} = BD^kC. \tag{3.2}$$

After $n + 1$ calls, the emissions are $x_0u, \dots, x_nu$ and the remainder is D^nCu. The three loop readings differ in how they combine this same exit sequence.

Closing the return port identifies its outgoing vector $Cu + Dy$ with its incoming vector y. Thus, for external input u, the connected wire must satisfy $y = Cu + Dy$. We call this the *wire equation*. Any solution gives external output $Au + By$.

The evaluator must compute the whole closed matrix, so it solves this equation on each external basis input s. Let y_s be a solution of the finite linear system $(I_Y - D)y_s = C|s\rangle$. The corresponding output column is $A|s\rangle + By_s$. Collecting the returning columns into a map $F : X \to Y$, with $F|s\rangle = y_s$, gives $(I_Y - D)F = C$ and output matrix $A + BF$. The rule first evaluates the open program, then solves its returning columns:

$$\frac{\kappa \Downarrow U \quad U = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) \quad F : X \to Y \quad (I_Y - D)F = C}{\mathrm{trace}_c\, \kappa \Downarrow A + BF}.$$

Applied to input u, the closed matrix $A + BF$ returns $(A + BF)u$ at type b, with the return summand removed. The earlier typing rules specify its boundary; this rule computes its action.

Theorem B.1 proves that a solution exists whenever the well-typed body κ evaluates to a finite unitary U. The matrix $I_Y - D$ may be singular, meaning it has no inverse. In that case two wire solutions can differ, but their difference v satisfies $Dv = v$. Unitarity then forces $Bv = 0$: a returning vector that preserves all its norm internally cannot also emit norm through B. Thus all solutions give the same external output, and the rule still determines one matrix.

To depict this connection, we use *sheet diagrams*, read upwards. Separate sheets carry sum alternatives, whereas wires at different depths on one sheet carry tensor factors [8, Defs. 16, 18–20]. The circled U at a seam acts on both incident summands; a sheet in front may hide part of the other

wire, but does not remove it. In Figure 3, bending the Y sheet back to that seam connects its output to its input.

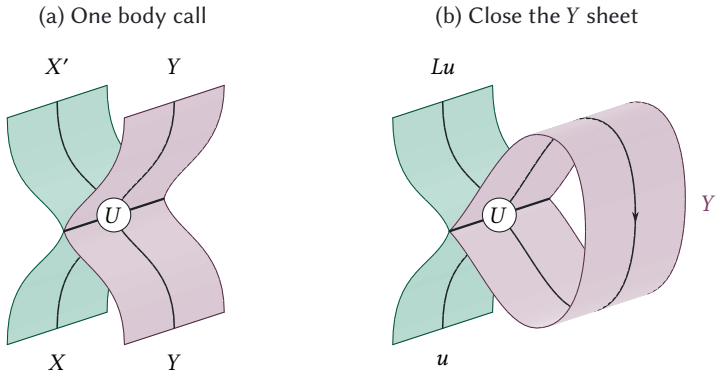


Fig. 3. Direct-sum feedback identifies the two ends of the returning Y sheet. The annotations evaluate the rotation by the trace rule.

For the rotation, elaboration supplies the atom $[R_\theta]$. Figure 3 solves its returning column, giving $F = 2$ and closed matrix (-1). The returning value 2 is a sum of amplitudes, rather than a normalized state on a new register. The computed closed iso is $\{* \leftrightarrow -1 \cdot *\} : 1 \leftrightarrow 1$, so

$$\texttt{trace guardRound on unit} \Downarrow -|*\rangle.$$

The closing operation removes both port tags and returns a value of type 1. A single call to the open `guardRound` still returns its tagged superposition from the first column.

Denotation. Coherent feedback adds all the emissions on their common external output:

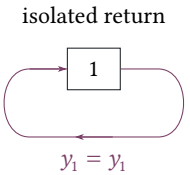
$$[\![\mathsf{trace}_c\ \beta]\!] = L = \sum_{k\geq 0} x_k.$$

This is the established direct-sum execution formula [1]. Theorem B.1 identifies it with the wire-equation output, so the operational rule and categorical trace $\mathrm{Tr}^Y(U)$ compute the same map.

Why is the sum defined for every finite body? For a finite unitary body, the established feedback theorem gives absolute convergence of the exits, a unitary sum L, and $\sum_k x_k^\dagger x_k = I_X$ [4, Thms. 4, 7, 8] [3, Thm. 3.3.15, Cor. 3.3.16]. Its proof separates the internal vectors that never emit from those that do. Unitarity prevents the external input from entering that closed component; on the emitting part the returns decay geometrically. Appendix B proves this by the norm accounting in Figure 16. No last nonzero exit is required.

Take $U = R_\theta \oplus 1$, with ordered coordinates (u, y_0, y_1). Here y_0 is the emitting return and y_1 is the isolated one. Connecting these two return ports gives

$$\tfrac{2}{5}y_0 = \tfrac{4}{5}u, \qquad y_1 = y_1, \qquad y_0 = 2u.$$



The isolated amplitude y_1 is arbitrary, but only y_0 enters the output $(3/5)u - (4/5)y_0 = -u$. Thus $D = \mathrm{diag}(3/5, 1)$ makes $I_Y - D$ singular without making the loop's output ambiguous. For rotations approaching the identity the loop remains -1, but at the identity it is $+1$ [9, Remark B.1].

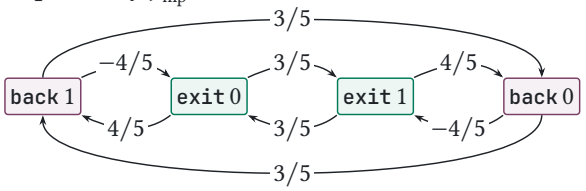
Computing the closed iso. The wire equation $y = Cu + Dy$ asks which amplitude must enter each return pattern when its output is connected back to its input. For a basis value $q \in \mathcal{B}(c)$, $y_q = \langle q|y\rangle$ is the amplitude on `back` q. The entry D_{qq} is the coefficient of `back` q in the ground clause whose input is also `back` q. For `guardRound`, $q = \star$ and $D_{qq} = 3/5$. The coordinate equation is

$$(1 - D_{qq})y_q = (Cu)_q + \sum_{r \neq q} D_{qr} y_r.$$

If $D_{qq} \neq 1$, divide by $1 - D_{qq}$ and substitute for y_q in the other return equations and the external output $v = Au + By$. In the ground clause table, this removes the input pattern `back` q and replaces every output occurrence of that pattern by the other terms of its clause, scaled by $1/(1 - D_{qq})$. Coefficients multiply along a substituted path and add when paths reach the same output. If $D_{qq} = 1$, unitarity forces every other entry in that row and column to vanish, so the isolated pattern can simply be removed. The smaller body is again unitary. Repeating with its updated coefficients eventually leaves only external clauses. Removing their `exit` tags gives the atom $[L] : a \leftrightarrow b$ with output $v = Lu$. This is the established elimination for finite contractions, including unitaries [3, Thm. 3.3.15]; Appendix B.1 proves order independence and preservation of the coefficient subfield.

Figure 4 follows two such steps for `flipRound`, whose body β_{flip} rotates the guard as before and flips a data bit on each call. Here u_0, u_1 are the input amplitudes on `exit` 0, 1, y_0, y_1 are the returning amplitudes on `back` 0, 1, and v_0, v_1 are the external output amplitudes. Substituting for y_0 creates a self-coefficient $9/25$ on the remaining return y_1. Solving that wire equation and substituting into the outputs gives $v = -u$. Appendix F.1 gives the iso clauses.

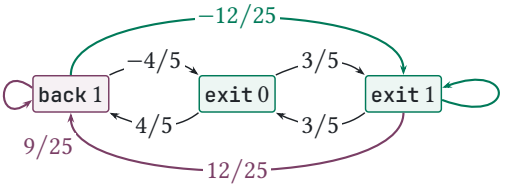
The open body β_{flip}



$$v = Au + By, \qquad y = Cu + Dy$$

$$v_0 = \tfrac{3}{5}u_1 - \tfrac{4}{5}y_1$$
$$v_1 = \tfrac{3}{5}u_0 - \tfrac{4}{5}y_0$$
$$y_0 = \tfrac{4}{5}u_1 + \tfrac{3}{5}y_1$$
$$y_1 = \tfrac{4}{5}u_0 + \tfrac{3}{5}y_0$$

1 Close `back` 0**: substitute for** y_0**.**



New terms from this substitution:

$$y_1 : \quad \tfrac{3}{5}\tfrac{3}{5}\, y_1 = \tfrac{9}{25} y_1$$
$$\tfrac{3}{5}\tfrac{4}{5}\, u_1 = \tfrac{12}{25} u_1$$
$$v_1 : \quad -\tfrac{4}{5}\tfrac{3}{5}\, y_1 = -\tfrac{12}{25} y_1$$
$$-\tfrac{4}{5}\tfrac{4}{5}\, u_1 = -\tfrac{16}{25} u_1$$
$$\left(1 - \tfrac{9}{25}\right) y_1 = \tfrac{4}{5}u_0 + \tfrac{12}{25}u_1$$
$$y_1 = \tfrac{5}{4}u_0 + \tfrac{3}{4}u_1$$

2 Close `back` 1**: solve for** y_1**.**



$$v_0 = \tfrac{3}{5}u_1 - \tfrac{4}{5}\left(\tfrac{5}{4}u_0 + \tfrac{3}{4}u_1\right) = -u_0 + \left(\tfrac{3}{5} - \tfrac{3}{5}\right)u_1 = -u_0$$
$$v_1 = \tfrac{3}{5}u_0 - \tfrac{16}{25}u_1 - \tfrac{12}{25}\left(\tfrac{5}{4}u_0 + \tfrac{3}{4}u_1\right) = \left(\tfrac{3}{5} - \tfrac{3}{5}\right)u_0 - \left(\tfrac{16}{25} + \tfrac{9}{25}\right)u_1 = -u_1$$

$$\text{trace}_{\text{Qubit}}\, \beta_{\text{flip}} : \text{Qubit} \leftrightarrow \text{Qubit}, \qquad v = Lu = -u, \qquad L = -I_2$$

Fig. 4. Eliminating the two returns of `flipRound`. An arrow gives an amplitude from its input port to its output port. The colored arrows in the middle graph are created by substituting for y_0; their products are at right. Solving the remaining equation gives the untagged output $v = -u$.

The rotation and its reference. For `guardRound`, put $t = 3/5$ and $s = 4/5$, so $A = D = t$, $C = s$ and $B = -s$. On input $|*\rangle$, the first call emits amplitude t and sends amplitude s back. Let r_n be the amplitude still returning after call n, with $r_1 = s$. The next call sends $-sr_n$ out and tr_n back. Coherent feedback adds the new emission to the sum ℓ_n of all earlier emissions. If the guard is measured on each call, we instead add the squared magnitude of that emission to the cumulative halting probability p_n. Starting with $\ell_1 = t$ and $p_1 = t^2$, these updates are

$$r_{n+1} = tr_n, \qquad \ell_{n+1} = \ell_n - sr_n, \qquad p_{n+1} = p_n + s^2|r_n|^2.$$

Table 1 separates the two calculations. The coherent sum can shrink through cancellation, while p_n only increases. Section 4 defines the recording and measured loops.

	one call		coherent feedback		guard measured
call n	emitted x_{n-1}	still returning r_n	sum emitted ℓ_n	squared length $\lvert\ell_n\rvert^2$	halted so far p_n
1	$\frac{3}{5}$	$\frac{4}{5}$	$\frac{3}{5}$	0.3600	0.3600
2	$-\frac{16}{25}$	$\frac{12}{25}$	$-\frac{1}{25}$	0.0016	0.7696
3	$-\frac{48}{125}$	$\frac{36}{125}$	$-\frac{53}{125}$	0.1798	0.9171
4	$-\frac{144}{625}$	$\frac{108}{625}$	$-\frac{409}{625}$	0.4282	0.9701
5	$-\frac{432}{3125}$	$\frac{324}{3125}$	$-\frac{2477}{3125}$	0.6283	0.9893
$n \to \infty$	0	0	-1	1	1

Table 1. `guardRound` on $|*\rangle$. Coherent feedback adds emissions, while guard measurement adds their squared magnitudes. A record keeps emission x_{n-1} with label $n-1$. Only p_n is a halting probability; decimal entries are rounded to four places.

In the limit $\ell_n \to -1$ and $p_n \to 1$, so both readings return the only external value $*$ with certainty. The coherent phase $-|*\rangle$ is global on this interface. To observe it, we place the loop beside an unaffected amplitude.

Add the bypass by changing the external type to `Qubit` = 1 + 1, with 0 for the bypass and 1 for the loop arm. Keep the return type `Unit`, leave external 0 unchanged, and rotate external 1 with the return port:

```
round withReference : Qubit + Unit = {
  exit 0 ↔ exit 0;
  exit 1 ↔ [ 3/5 . exit 1 + 4/5 . back unit];
  back unit ↔ [-4/5 . exit 1 + 3/5 . back unit]
}
```

Use `exit 0`, `exit 1`, `back unit` as the ordered input and output bases. The three clauses give the three columns below. The first two rows and columns belong to the external qubit, and the last row and column belong to the return port. The boxed letters name the four blocks of (3.1).

$$U_{\text{ref}} = \begin{array}{c|ccc} & \texttt{exit}\,0 & \texttt{exit}\,1 & \texttt{back}\,* \\ \hline \texttt{exit}\,0 & 1 & 0 & 0 \\ \texttt{exit}\,1 & 0 & t & -s \\ \texttt{back}\,* & 0 & s & t \end{array} \qquad \begin{array}{l} A = \operatorname{diag}(1,t) \\ B = \begin{pmatrix} 0 \\ -s \end{pmatrix} \\ C = \begin{pmatrix} 0 & s \end{pmatrix} \\ D = t \end{array}$$

Substituting these blocks into the exit formula gives

$$x_0 = \operatorname{diag}(1,t), \qquad x_{k+1} = \begin{pmatrix} 0 \\ -s \end{pmatrix} t^k \begin{pmatrix} 0 & s \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & -s^2 t^k \end{pmatrix}.$$

Only the immediate exit contains the bypass. Summing the lower right entries gives -1 as before, so the closed gate is $L = \operatorname{diag}(1,-1)$ on the external qubit. Finite elimination gives the same external clauses: $\{0 \leftrightarrow 0,\ 1 \leftrightarrow -1 \cdot 1\}$.

The standard *Hadamard test* makes this relative phase observable [7, p. 1, Eqs. (1)–(3)]. Its Hadamard iso has two clauses, one for each input column:

$$H = \{0 \leftrightarrow \tfrac{1}{\sqrt{2}}(0+1),\quad 1 \leftrightarrow \tfrac{1}{\sqrt{2}}(0-1)\} : \texttt{Qubit} \leftrightarrow \texttt{Qubit}, \qquad [\![H]\!] = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

It sends input $|0\rangle$ to $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, supplying amplitude to both arms. The coherent loop changes this to $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$. Applying H again gives $|1\rangle$, so a computational-basis measurement returns 1 with certainty. The composition $H; \mathsf{trace}_1\ \mathsf{withReference}; H$ uses the Hadamard iso just defined on both sides of the closed body. Here the controlled operation is just the scalar loop, so the control qubit is the entire external data. Figure 5 shows the test and expands its two arms. Section 4 will run the same test after measuring the loop's return tag and obtain probability $1/5$ for outcome 1.

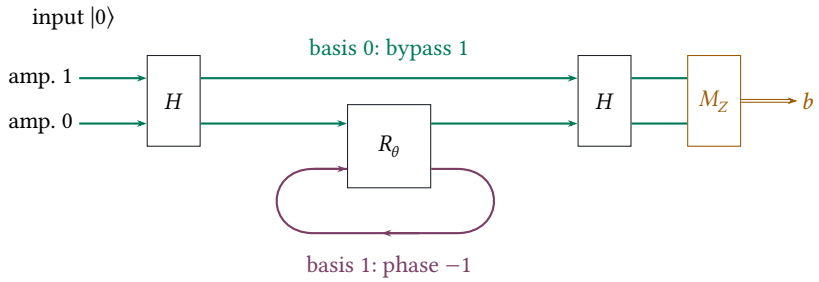


Fig. 5. The two rails are the basis alternatives of $\texttt{Qubit} = 1 + 1$. Only basis 1 enters the scalar loop, whose rotation has $\cos\theta = t = 3/5$. The second H recombines its phase with the bypass, giving $b = 1$ with certainty.

Elimination computes L without retaining its decomposition into exits. To compare feedback with measurement, we now keep that decomposition: attach a count label to each $x_k u$, discard those labels and compare the returned data with Lu.

4 What allocation and hiding do to the loop

Returning a count requires a register the caller did not supply. Following *Quantum Information Effects*, allocation exposes an input summand of a unitary and hiding discards an output factor of an

isometry [11, §§ 3–4, Thm. 22]. These extend finite iso first to iso_a isometries, then to iso_a^{χ} channels. We need both extensions to carry out the classical idea from the introduction: allocation supplies the count register, and hiding lets the caller ignore it. Together they also derive measurement, which will distinguish exit from return inside every call.

Syntax. An allocation program $v : a \rightarrowtail b$ is built from $\mathsf{arr}\,\omega$, $\mathsf{lift}\,\omega$, sequence, sum and tensor; a channel program $h : a \rightsquigarrow b$ uses $\mathsf{arr}\,v$, $\mathsf{hide}\,v$, sequence and tensor. We add the forms $\mathsf{count}_K\,\omega$ and $\mathsf{while}_c\,\omega$, with $K \geq 1$ and a reversible body. Their definitions follow below; the full grammars and inherited rules are in Appendix A. The executable spelling `while f` infers c from a named `round`, as `trace f` does.

Statics. An embedding arr retains its program's input and output types. Given $\omega : a + d \leftrightarrow b$, allocation produces $\mathsf{lift}\,\omega : a \rightarrowtail b$ by selecting the a input summand. Given $v : a \rightarrowtail b \otimes g$ with $\mathscr{B}(g) \neq \emptyset$, hiding produces $\mathsf{hide}\,v : a \rightsquigarrow b$ by removing the g factor. The loop rules, with their common environment Σ omitted, are

$$\frac{\beta : a + c \leftrightarrow b + c \qquad K \geq 1}{\mathsf{count}_K\,\beta : a \rightarrowtail [K] \otimes b} \qquad\qquad \frac{\beta : a + c \leftrightarrow b + c}{\mathsf{while}_c\,\beta : a \rightsquigarrow b}.$$

Sequence matches intermediate types, while tensor forms products of input and of output types. Isometries also combine by direct sum, while channels have no coherent sum constructor. The type $[K]$ is the K-fold sum of 1, with basis labels $0, \ldots, K-1$.

Dynamics. If $\omega \Downarrow U$, evaluating $\mathsf{lift}\,\omega$ uses only the input columns belonging to a, so input u gives $U(u, 0)$. If $v \Downarrow V$, evaluating $\mathsf{hide}\,v$ on a density operator ρ gives $\mathrm{tr}_g(V\rho V^\dagger)$. Here a density operator is a positive semidefinite matrix of trace one; a unit vector u has state $|u\rangle\langle u|$. The partial trace sums diagonal blocks indexed by the hidden factor's basis. The first arr retains the unitary matrix, while the second sends an isometry V to $\mathscr{E}_V : \rho \mapsto V\rho V^\dagger$. The loop evaluations below use the same finite exit matrices.

Denotation. The allocation and hiding completions turn finite unitaries $\mathbf{U}$ into isometries $\mathbf{I} \simeq R[\mathbf{U}]$ and then channels $\mathbf{Q} \simeq L[\mathbf{I}]$ [11, Prop. 7, Thm. 22]. They expose an input summand and hide an output factor as in the evaluation rules above. Appendix A gives the representatives and their defining equivalence relations.

Allocation, discard and measurement. The operations we need can now be defined from these constructors. For $0 \leq j < K$, put $d = [K-1] \otimes a$, with $[0] = 0$, and let $p_j : a + d \leftrightarrow [K] \otimes a$ be a basis permutation that places the exposed a summand at label j. Let $\epsilon_b : b \leftrightarrow 1 \otimes b$ insert the unit factor, and let $\mathsf{clone}_b : b \rightarrowtail b \otimes b$ send each basis value s to $\langle s, s\rangle$. This last map is a lift of a permutation onto the diagonal basis values. Define allocation for any a, and discard and measurement for inhabited b:

$$\begin{aligned} \mathsf{alloc}_j &= \mathsf{lift}\,p_j && : a \rightarrowtail [K] \otimes a, \\ \mathsf{discard}_b &= \mathsf{hide}(\mathsf{arr}\,\epsilon_b) && : b \rightsquigarrow 1, \\ \mathsf{measure}_b &= \mathsf{arr}\,\mathsf{clone}_b\,;(I_b \otimes \mathsf{discard}_b)\,;\,\mathsf{unit}_{\otimes} && : b \rightsquigarrow b. \end{aligned}$$

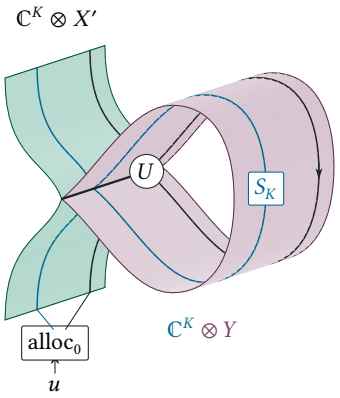
Here identities and $\mathsf{unit}_{\otimes} : b \otimes 1 \leftrightarrow b$ are embedded as channels. Allocation sends u to $|j\rangle \otimes u$; discard sends ρ to its trace. Cloning copies the chosen *basis values*, so a superposition becomes $\sum_s \alpha_s |s\rangle \otimes |s\rangle$. Hiding its second factor removes the off-diagonal terms, giving

$$[\![\mathsf{measure}_b]\!](\rho) = \sum_{s \in \mathscr{B}(b)} P_s \rho P_s, \qquad P_s = |s\rangle\langle s|.$$

This is the derived computational-basis measurement of *Quantum Information Effects* [11, § 3.4.1]. It retains the measured value in its output and forgets the copy. These operations supply the counter and the measurement for one body call. Closing the return uses coherent feedback for `count` and classical iteration after measurement for `while`.

4.1 Keep a count

Allocate a register $[K]$ at 0, with $K \geq 1$, and carry it through the body. On a return, advance it by the reversible successor $S_K : j \mapsto j+1 \bmod K$. Distributivity identifies $\mathbb{C}^K \otimes (X \oplus Y)$ with $(\mathbb{C}^K \otimes X) \oplus (\mathbb{C}^K \otimes Y)$, so each alternative sheet carries counter and payload together [8, § 6]. Let $d_K : [K] \otimes (a+c) \leftrightarrow ([K] \otimes a) + ([K] \otimes c)$ be distributivity, and let d'_K distribute over $b+c$. Figure 6 defines the enlarged body β_K and closes its return before supplying the allocated input. Its matrix W_K carries the counter on both alternatives and advances it only on a return.



The enlarged body and the count program are

$$\begin{aligned} \beta_K &= d_K^{-1} ; (I_{[K]} \otimes \beta) ; d'_K \\ &\quad ; (I_{[K]\otimes b} \oplus (S_K \otimes I_c)), \\ \mathrm{count}_K\, \beta &= \mathrm{alloc}_0 ; \mathrm{arr}(\mathrm{trace}_{[K]\otimes c}\, \beta_K). \end{aligned}$$

Their matrices satisfy

$$\begin{aligned} W_K = [\![\beta_K]\!] &= \begin{pmatrix} I_K \otimes A & I_K \otimes B \\ S_K \otimes C & S_K \otimes D \end{pmatrix}. \\ R_K u &= (\mathrm{Tr}^{\mathbb{C}^K \otimes Y}\, W_K)(|0\rangle \otimes u) \\ &= \sum_{k \geq 0} |k \bmod K\rangle \otimes x_k u. \end{aligned} \tag{4.1}$$

Fig. 6. The count construction. Counter and payload are distinct tensor wires on both sheets, shown in depth in Figure 7(a). Only the returning counter passes through S_K.

The first call of W_K on $(|0\rangle \otimes u, 0)$ emits $|0\rangle \otimes Au$ and returns $|1 \bmod K\rangle \otimes Cu$. Here $|0\rangle$ is a prepared counter, while the bare 0 is absent amplitude. Closing $\mathbb{C}^K \otimes Y$ adds the exits, with the counter advanced once on each return. Figure 7 follows three calls with $K > 3$. The blue counter and black payload travel together on each sheet. The right panel closes the original body and adds its exits without count labels.

To hide the counter, first use the symmetry iso $\sigma_{[K],b} : [K] \otimes b \leftrightarrow b \otimes [K]$. Hiding then gives $\mathsf{hide}(\mathrm{count}_K \beta; \mathrm{arr}\, \sigma_{[K],b})$. Native `count` returns data first, so `discard` hides its counter directly. Counts congruent modulo K share a label, so their amplitudes interfere before the clock is discarded. Hiding this finite clock therefore need not give the measured loop.

Allocation is an isometry and the enlarged coherent loop is unitary, so their composite R_K is an isometry. Evaluation returns the joint state without sampling a halting round.

An exact count uses $H_{\mathrm{rec}} = \ell^2(\mathbb{N})$ and $Ru = \sum_k |k\rangle \otimes x_k u$, with $\|Ru\|^2 = \sum_k \|x_k u\|^2 = \|u\|^2$. It is the semantic limit of finite clocks. For the scalar rotation, $R|*\rangle = \xi_{\mathrm{rot}} \otimes |*\rangle$ with unit count state $\xi_{\mathrm{rot}} = t|0\rangle - s^2 \sum_{k\geq 1} t^{k-1}|k\rangle$. For `withReference`, the bypass instead has record $|0\rangle$: $R|0\rangle = |0\rangle \otimes |0\rangle$, while $R|1\rangle = \xi_{\mathrm{rot}} \otimes |1\rangle$. These different records of the two arms make $R|+\rangle$ entangled.

For `guardRound` at $K = 2$, the even-count amplitudes sum to $t - s^2 t/(1-t^2) = 0$, and the odd-count amplitudes sum to $-s^2/(1-t^2) = -1$. Its parity clock is therefore $-|1\rangle$. The bypass retains label $|0\rangle$.

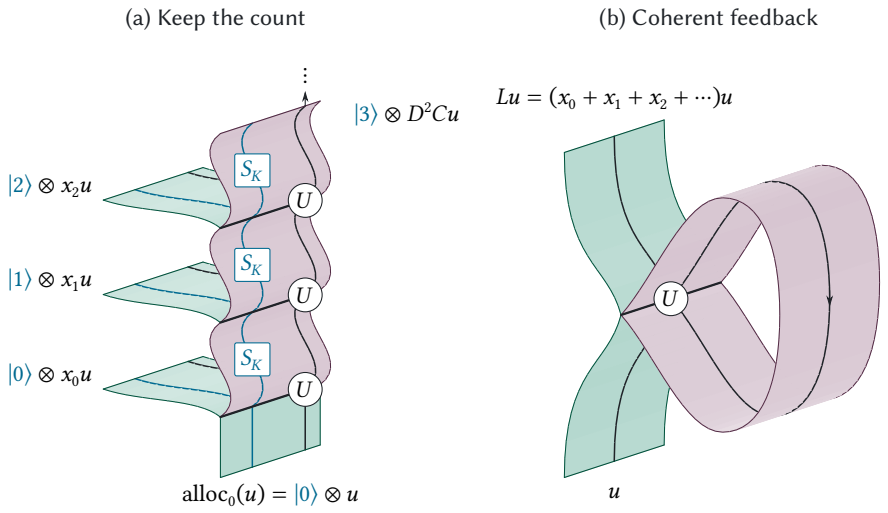


Fig. 7. After three calls of a $K > 3$ clock, $|3\rangle \otimes D^2Cu$ continues past the arrow and dots. The right panel closes the original body.

The parity clock therefore distinguishes the two arms. The joint output on $|+\rangle$ is $(|0\rangle|0\rangle - |1\rangle|1\rangle)/\sqrt{2}$, so tracing out the clock leaves $I/2$.

Including each finite counter in H_{rec}, these outputs converge in operator norm to the exact record R (Appendix C). A time-dependent model can keep the same distinctions by making outputs from different calls arrive at different times [3, § 3.4]. Synchronizing execution has a different purpose. The quantum control machine synchronizes final control states [22], and register-machine compilation obtains wait counts within a chosen time bound [23, § 5]. These address common-clock halting [16] without requiring the program to return a count register.

4.2 Measure the returning tag

To measure the guard without measuring the payload, allocate a fresh bit at 0 and flip it only on a `back` tag. Hiding this copied bit removes the cross terms between exit and return, while preserving coherence within either payload. This works for unequal payloads, as in `withReference`. The tag-copying isometry is

$$T : b + c \rightarrowtail (b + c) \otimes \texttt{Qubit}, \qquad \mathsf{exit}\, x \mapsto \langle \mathsf{exit}\, x, 0\rangle, \quad \mathsf{back}\, y \mapsto \langle \mathsf{back}\, y, 1\rangle.$$

The full reversible clauses are in Appendix A. One call is $\mathsf{hide}(\mathsf{arr}\,\beta; T)$, which keeps the payload and removes only coherence between the two tags.

For a density input ρ of trace one, the first call emits $A\rho A^\dagger$ and returns $\sigma_1 = C\rho C^\dagger$. Subsequent calls emit $B\sigma_n B^\dagger$ and return $\sigma_{n+1} = D\sigma_n D^\dagger$. The blocks are not normalized after each call: $\mathrm{tr}\,\sigma_n$ is precisely the probability of still being inside after n calls. The emitted block on call $k+1$ is $x_k \rho x_k^\dagger$. Adding these emitted blocks gives the measured loop $\mathscr{M} = [\![\mathsf{while}_c\ \beta]\!]$:

$$\mathscr{M}(\rho) = \sum_{k \geq 0} x_k \rho x_k^\dagger = \mathrm{tr}_{H_{\mathrm{rec}}}(R\rho R^\dagger). \tag{4.2}$$

The residual probability tends to zero by Theorem B.1. Thus a measured run halts after finitely many calls with probability one, although there need not be a finite bound on its possible halting

rounds. Its output is a density operator on $[\![b]\!]$, with no port tag or retained count. This differs from coherent evaluation, which solves for a completed map without sampling a halting round.

For `withReference` on $\rho = |+\rangle\langle+|$, the immediate exit is the only one containing both paths. Its density matrix and the sum of all later contributions are

$$\underbrace{\frac{1}{2}\begin{pmatrix}1 & t\\ t & t^2\end{pmatrix}}_{\text{immediate exit}} + \underbrace{\frac{s^4}{2(1-t^2)}\begin{pmatrix}0 & 0\\ 0 & 1\end{pmatrix}}_{\text{later exits}} = \frac{1}{2}\begin{pmatrix}1 & t\\ t & 1\end{pmatrix}.$$

Here $s^4/(1-t^2) = s^2$, so the lower diagonal entry is $(t^2+s^2)/2 = 1/2$. A final Hadamard followed by reading 1 tests the projection onto $|-\rangle$. Its measured probability is $(1-t)/2 = 1/5$, whereas the coherent output $|-\rangle$ gives probability one. Both predictions therefore follow from the same three clauses.

Discarding the exact record gives the same mixture in (4.2): tracing out $|k\rangle\langle j|$ removes $k \neq j$ and keeps $x_k \rho x_k^\dagger$ at $k = j$.

For general payloads, let (ρ, σ) be positive blocks on X, Y with total trace at most one. Applying U to their block-diagonal matrix and retaining the two diagonal output blocks gives the branching map

$$\Phi_U(\rho,\sigma) = (A\rho A^\dagger + B\sigma B^\dagger, \quad C\rho C^\dagger + D\sigma D^\dagger).$$

Its two outputs are a classical alternative: their traces are the weights on the exit and return branches. When $a = b = c$, factor $: a + a \leftrightarrow a \otimes \texttt{Qubit}$ exposes the tag. Its inverse dist restores the sum, so the left box in Figure 8 is $\widehat{U} = [\![\mathsf{dist}; \beta; \mathsf{factor}]\!]$.

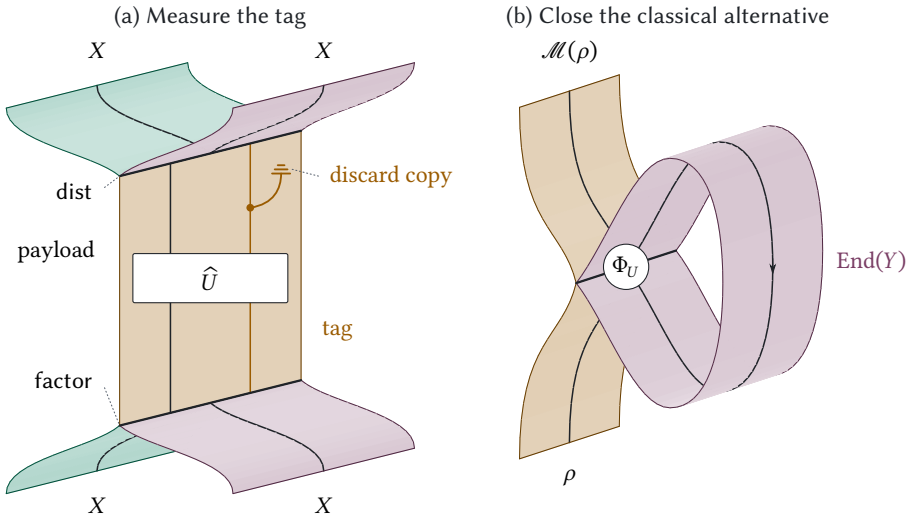


Fig. 8. At left (for $X = X' = Y$), the copied tag is hidden inside each call, before the outgoing sheets separate. At right, closing the returning classical block of Φ_U gives the measured loop for general payloads.

Closing the return uses the established trace of completely positive trace-nonincreasing maps on finite C^*-algebras, with their classical sum [3, Prop. 3.2.17]:

$$[\![\mathsf{while}_c\ \beta]\!] = \mathrm{Tr}_{\mathrm{cl}}^{\mathrm{End}(Y)}(\Phi_U) = \mathscr{M}.$$

Unrolling this trace gives the emitted blocks in (4.2).

Our body-first convention is obtained by applying U once before the published test-first loop [21, § 3]. Appendix C gives the correspondence.

Orthogonal exits in retained data can still have nonzero cross outer products. An ignored count removes those terms; the two-exit cycle in Appendix F illustrates the distinction.

Allocation alone preserves coherent feedback [3, Cor. 3.3.16]. Recording changes the body by adding a counter and successor. Measurement instead closes a classical alternative of operator blocks.

Definition 4.1 (The three loop readings). Let $\beta : a + c \leftrightarrow b + c$ be finite, with exits x_k and $K \geq 1$. The right column specializes to `withReference` on $u = |+\rangle$ or $\rho = |+\rangle\langle+|$, with $K = 2$ and $t = 3/5$.

$\mathsf{trace}_c\ \beta : a \leftrightarrow b$	$Lu = \sum_{k\geq 0} x_k u$	$\lvert-\rangle$
$\mathsf{count}_K\ \beta : a \rightarrowtail [K] \otimes b$	$R_K u = \sum_{k\geq 0} \lvert k \bmod K\rangle \otimes x_k u$	$\dfrac{\lvert0\rangle\lvert0\rangle - \lvert1\rangle\lvert1\rangle}{\sqrt{2}}$
$\mathsf{while}_c\ \beta : a \rightsquigarrow b$	$\mathscr{M}(\rho) = \sum_{k\geq 0} x_k \rho x_k^\dagger$	$\dfrac{1}{2}\begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}$

5 When can a caller forget the difference?

The Hadamard test distinguished the measured and coherent loops using one output bit. We now ask when their channels are equal on *every* input, including data entangled with a caller's reference. We call this *free erasure* of the count. The unitary-channel criterion is the rank-one case of uniqueness of finite Kraus representations [20, Cor. 2.23]. Here its Kraus operators are the body's successive exits. Their shared finite return space makes the criterion a finite test on clauses. The scalar rotation gives a first case of agreement: its count state is fixed, so ignoring it cannot change the data. The following criterion determines when the same is true for a general external interface.

Theorem 5.1 (Agreement on the data). *Let $\beta : a + c \leftrightarrow b + c$ be a finite iso with nonempty external type. Put $m = |\mathscr{B}(c)|$. The following conditions are equivalent.*

(1) $\mathscr{M} = \mathscr{E}_L$ as channels on the external data, where $\mathscr{E}_L(\rho) = L\rho L^\dagger$.

(2) There is a single unit vector $\xi \in \ell^2(\mathbb{N})$ such that $Ru = \xi \otimes Lu$ for every u.

(3) The $m + 1$ operators $x_0, \ldots, x_m$ span a one-dimensional subspace in the space of maps from X to X'.

In this case $x_k = b_k L$ for every k, with $\sum_k |b_k|^2 = 1$ and $\sum_k b_k = 1$.

For the scalar rotation, $Ru = \xi_{\mathrm{rot}} \otimes u = (-\xi_{\mathrm{rot}}) \otimes Lu$ since $Lu = -u$, so the fixed state in condition (2) is $-\xi_{\mathrm{rot}}$. For `withReference`, the return space still has dimension $m = 1$, but the first two exits are

$$x_0 = \mathrm{diag}(1, 3/5), \qquad x_1 = \mathrm{diag}(0, -16/25).$$

They cannot be multiples of one another, since the first is identity on the bypass and the second is zero there. Thus the first two exits already certify the difference observed by the Hadamard test in Section 4. Agreement also permits a loop to change its data. Applying X only on exit makes every exit a scalar multiple of X and their sum −X, so this variant passes. Appendix F gives its clauses and a two-section cascade that also passes on non-scalar data.

Agreement concerns the specified interface. A channel identifies I and $-I$, whereas a coherent bypass distinguishes $I \oplus I$ from $I \oplus (-I)$. Such a bypass therefore has to be formed before channelization, which preserves neither that coherent sum nor its trace for arbitrary bodies [3, § 3.3.3].

The rank-one Kraus criterion gives $x_k = b_k L$, so $Ru = \xi \otimes Lu$ and tracing out the fixed count preserves $L\rho L^\dagger$. Cayley–Hamilton then puts every later exit in the span of the first $m + 1$; at $m = 0$, only $x_0 = L$ remains. Exact coefficients with decidable equality give the finite test, using any nonzero exit as reference. Appendix D proves the equivalences.

Figure 9 puts the two constructions beside one another. Write $\mathrm{End}(X)$ for the matrix algebra on X. The lower path uses the classical sum $\mathrm{End}(X) \oplus \mathrm{End}(Y)$, whose inputs are pairs of positive blocks. It differs from the coherent Hilbert sum $X \oplus Y$ used on the upper path. The theorem says exactly when closing these different kinds of return gives the same channel on the external data.

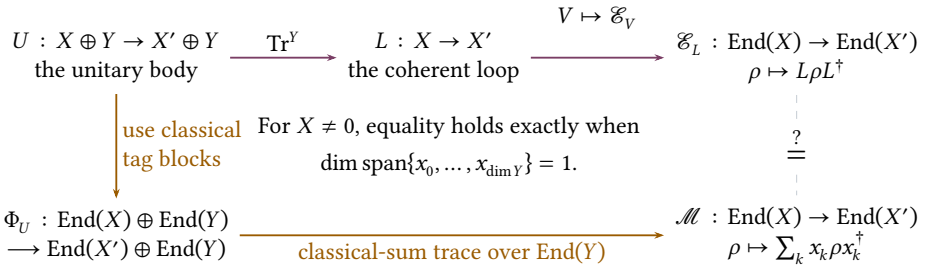


Fig. 9. Two constructions from one body. The upper path closes amplitudes before taking a data channel. The lower path uses classical tag alternatives and then closes their return. Theorem 5.1 decides whether the two resulting channels coincide.

A program transformation. The Hadamard caller compares the loops as channels, using the syntactic embedding $E(\omega) = \mathrm{arr}(\mathrm{arr}\,\omega)$, which denotes $\mathscr{E}_{[\![\omega]\!]}$. Let $\mathrm{prep}_+ : 1 \rightsquigarrow \texttt{Qubit}$ allocate 0, remove the unit factor and apply H, and let $\mathrm{read}_Z : \texttt{Qubit} \rightsquigarrow \texttt{Qubit}$ be the derived basis measurement, whose output records the classical outcome. The context

$$\mathscr{C}[-] = \mathrm{prep}_+ ; [-] ; E(H) ; \mathrm{read}_Z$$

has hole type $\texttt{Qubit} \rightsquigarrow \texttt{Qubit}$ and result type $1 \rightsquigarrow \texttt{Qubit}$. Plugging in $E(\texttt{trace withReference})$ gives outcome 1 with probability one, while $\texttt{while withReference}$ gives it with probability $1/5$.

The permitted observers are channel contexts with one hole of type $a \rightsquigarrow b$. For fixed closed channels h, h_j, their grammar is

$$\mathscr{C} ::= [-] \mid h; \mathscr{C} \mid \mathscr{C}; h \mid h \otimes \mathscr{C} \mid \mathscr{C} \otimes h$$
$$\mid \mathrm{case}\big(\Psi_i; \mathscr{C}, (\Psi_j; h_j)_{j \neq i}\big).$$

The last constructor uses an instrument with a finite outcome set J and a chosen branch $i \in J$. Its completely positive maps $\Psi_j : \mathrm{End}([\![c]\!]) \to \mathrm{End}([\![c_j]\!])$ have output traces summing to one on a density input. The trace of $\Psi_j(\rho)$ is the probability of outcome j. Branch i has payload type c_i, which the selected context accepts; the other continuations accept their respective c_j, and all branches return the same final type. The judgment $a \rightsquigarrow b \vdash \mathscr{C} : c \rightsquigarrow d$ means that plugging a program of the hole type gives a channel $c \rightsquigarrow d$. Sequence matches intermediate types, while tensor pairs the two inputs and the two outputs. The full rules are in Appendix D. Branching uses classical outcomes, so these contexts supply no coherent control or coherent trace of the channel hole.

More generally, on density input ρ an output test $0 \leq P \leq I$ has probability $\mathrm{tr}(P\,[\![\mathscr{C}[h]]\!](\rho))$. Two channel programs are *contextually equivalent* when these probabilities agree for every well-typed context, input and test. Equality of channels preserves them by induction on the grammar, including an entangled reference in a tensor factor. Conversely, unequal channels have a distinguishing input and output test already in the empty context.

Corollary 5.2 (Replacement). *Dropping the exact record always gives* `while`. *On nonempty external data, an embedded* `trace` *may replace* `while` *in all channel contexts exactly when the test of Theorem 5.1 passes. Then dropping any finite clock gives the same channel. On empty external data all these data channels coincide.*

The exact record lies outside the finite grammar. Appendix D covers semantic continuations that carry it without inspection and drop it on return.

Reading a fixed count still adds an observable output. Even when $K = \dim Y$, a single clock can miss a failure of agreement. On an external qubit, a permutation can have only $x_0 = |0\rangle\langle 0|$ and $x_K = |1\rangle\langle 1|$ as nonzero exits. Both receive clock label 0, so discarding that clock preserves $|+\rangle$, whereas measuring each call gives $I/2$. Appendix D gives the permutation.

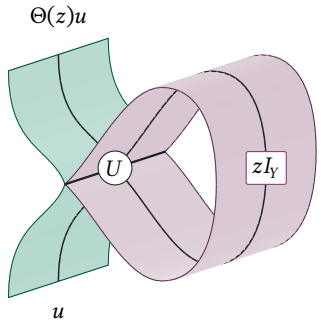
6 A count without a counter

The agreement test concerns channels on returned data. We now keep the coherent iso before embedding it as a channel, so we can put it beside the reference used in Section 3. Changing the phase on its return wire will reveal how its response depends on the rounds.

A completed gate alone does not determine a count distribution: an immediate phase -1 and `guardRound` both give -1, but only `guardRound` has delayed exits. We distinguish them by inserting a phase $z = e^{i\varphi}$ on the return port and varying it. The new completed gate $\Theta(z)$ leaves an immediate exit unchanged and multiplies exit k by z^k, one factor for each return. This *transfer function* therefore generates the original exits. Summing the marked exits will use an inverse on the returning space, so we first separate the vectors that never emit. Define

$$Y_0 = \{y \in Y \mid BD^j y = 0 \text{ for every } j \geq 0\}, \qquad Y_1 = Y_0^{\perp}.$$

By definition, $D(Y_0) \subseteq Y_0$ and $B|_{Y_0} = 0$. Unitarity gives $\|Dy\|^2 + \|By\|^2 = \|y\|^2$, so D preserves norm on Y_0. In finite dimension this restriction is onto. Unitarity then maps the orthogonal complements of these internal components onto one another, giving $C(X) \subseteq Y_1$ and $D(Y_1) \subseteq Y_1$. Put $B_1 = B|_{Y_1}$ and $D_1 = D|_{Y_1}$, regarding C as $X \to Y_1$. The geometric decay from Section 3 makes $I - zD_1$ invertible for $|z| \leq 1$, even when $I - zD$ on all of Y is singular.



	$j=0$	$j=1$	$j=2$
$k=0$	T_{00}	$z^{-1}T_{01}$	$z^{-2}T_{02}$
$k=1$	zT_{10}	T_{11}	$z^{-1}T_{12}$
$k=2$	z^2T_{20}	zT_{21}	T_{22}

$$T_{kj} = x_k \rho x_j^{\dagger}$$

$$\Theta_{\mathrm{rot}}(z) = \frac{t - z}{1 - tz},$$

$$\Theta_{\mathrm{ref}}(z) = \mathrm{diag}(1, \Theta_{\mathrm{rot}}(z)).$$

$$\Theta(z) = \sum_{k \geq 0} z^k x_k = A + zB_1(I - zD_1)^{-1}C, \qquad L = \Theta(1).$$

Fig. 10. Each return adds a factor z at left. In the channel at right, exit pair k, j carries $z^{k-j}T_{kj}$. Averaging over the circle retains only the orange diagonal. The same cancellation applies to later exits. The displayed families Θ_{rot} and Θ_{ref} belong to `guardRound` and `withReference`.

For the data channel, the term $x_k \rho x_j^\dagger$ carries phase $e^{i(k-j)\varphi}$, as the right panel of Figure 10 shows. Averaging over the circle keeps only $k = j$, so

$$\mathscr{M}(\rho) = \frac{1}{2\pi} \int_0^{2\pi} \Theta(e^{i\varphi})\, \rho\, \Theta(e^{i\varphi})^\dagger \, d\varphi. \tag{6.1}$$

Thus the measured loop is a uniform mixture of coherent loops differing only in their return phase. Geometric decay justifies these Fourier manipulations.

Differentiation can recover a count because a single exit at count k has response $e^{ik\varphi}L$. Its derivative is ik times that response. Undoing the gate and multiplying by $-i$ gives kI. With several exits there are also cross terms at a fixed phase. The unitary transfer identity and the phase average from monitored recurrence [5, Eq. (4.10)] [6, § 3, Eq. (8)] relate this coherent observable to the measured mean.

Theorem 6.1 (The measured mean and the phase derivative). *Let $F_z = (I - zD_1)^{-1}C$ and $N = \sum_{k\ge 0} k\, x_k^\dagger x_k$. For $z = e^{i\varphi}$, the positive semidefinite relative phase derivative and its average satisfy*

$$Q(z) = -i\Theta(z)^\dagger \frac{d}{d\varphi}\Theta(e^{i\varphi}) = F_z^\dagger F_z,$$

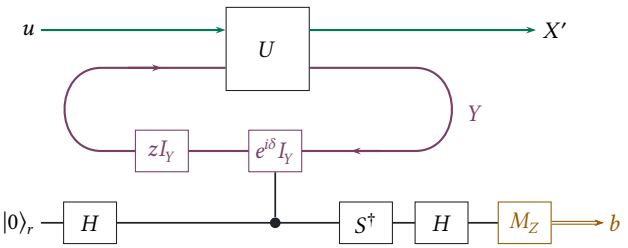
$$\frac{1}{2\pi}\int_0^{2\pi} Q(e^{i\varphi})\, d\varphi = N, \qquad \operatorname{tr} N = \dim Y_1.$$

For unit $u \in X$, $\langle u, Nu\rangle$ is the measured mean number of returns.

Appendix E differentiates U's inner-product identity and averages over phase. With $CC^\dagger = I_{Y_1} - D_1 D_1^\dagger$ from unitarity, cyclicity and decay give the telescope

$$\operatorname{tr} N = \sum_{j\ge 0} \operatorname{tr}\big(D_1^j C C^\dagger (D_1^\dagger)^j\big) = \lim_{n\to\infty} \operatorname{tr}\big(I_{Y_1} - D_1^n (D_1^\dagger)^n\big) = \dim Y_1.$$

For a unit input u, prepare a reference bit in $|+\rangle$ to control only the choice of return phase z or $ze^{i\delta}$. The $S^\dagger; H$ readout in Figure 11 measures the outputs' imaginary overlap, with inner products linear in the second argument [7, p. 1, Eqs. (1)–(3)]. Appendix I gives its pure iso construction. Since $Q(z) = F_z^\dagger F_z$, the first-order response gives the squared norm of the returning amplitude before the phase gate. Several rounds contribute to this wire, so its circulating weight can exceed one.



$U : X \oplus Y \to X' \oplus Y$

guardRound, $u = 1$

z	F_z	$Q(z)$
1	$s/(1-t) = 2$	4
-1	$s/(1+t) = 1/2$	1/4

$S^\dagger = \operatorname{diag}(1, -i)$

$$\Pr(0) - \Pr(1) = \operatorname{Im}\langle \Theta(z)u, \Theta(ze^{i\delta})u\rangle = \delta\langle u, Q(z)u\rangle + O(\delta^2).$$

Fig. 11. The reference chooses return phase z or $ze^{i\delta}$ and accompanies both summands unchanged through U. Its final measurement reads the relative derivative. At right, $t = 3/5$, $s = 4/5$, and F_z is the scalar return amplitude before the phase gate.

For guardRound, the measured mean is $\sum_{k\ge 1} k s^4 t^{2(k-1)} = \sum_{j\ge 0} s^2 t^{2j} = s^2/(1-t^2) = 1$. The second sum adds survival probabilities. The table beside the circuit instead squares the summed

return amplitudes, giving coherent weights 4 and $1/4$. Averaging $Q(z)$ over all phases recovers the measured mean.

The bypass in `withReference` makes no return, giving $N_{\mathrm{ref}} = \mathrm{diag}(0, 1)$. Its measured mean on $|+\rangle$ is therefore $1/2$, the equal-weight average of bypass count zero and scalar-loop mean one.

Putting return phase $z = e^{i\varphi}$ into the original Hadamard caller gives these observable probabilities:

$$\Pr(b = 1) = |1 - \Theta_{\mathrm{rot}}(z)|^2/4$$

φ	0	$\pi/2$	π
$\Theta_{\mathrm{rot}}(e^{i\varphi})$	-1	$(15 - 8i)/17$	1
$\Pr(b = 1)$	1	$1/17$	0

Averaging this test over the phase circle gives the measured probability $1/5$ by (6.1).

On a nonempty external type, agreement gives $\Theta(z) = f(z)L$ with $|f(z)| = 1$ on the circle, so every return phase gives the same data channel. Conversely, if every phase channel equals $\mathscr{C}_L$, their average is both $\mathscr{C}_L$ and $\mathscr{M}$ by (6.1). Thus the finite test also decides whether changing the return phase can affect the data.

Appendix H realizes the scalar response with an ideal lossless ring coupler [2, 18].

7 The three readings in the Π family

Figure 12 places the three readings in the Π family. A backwards wire is supplied by the standard Int construction: its objects are pairs (P^+, P^-), and a map to (Q^+, Q^-) is a unitary $P^+ \oplus Q^- \to Q^+ \oplus P^-$. The fully faithful embedding $N(X) = (X, 0)$ preserves the sum trace [13, 17]. Appendix G draws the typed components of this bending.

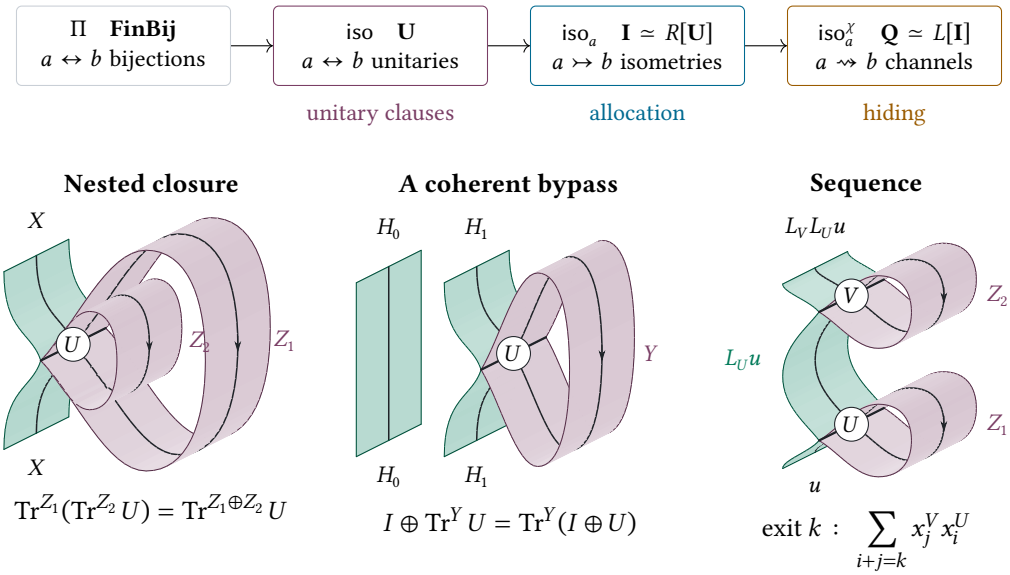


Fig. 12. Finite source extensions and coherent trace laws for endomorphisms [4, 11, 13]. Here $L = \sum_k x_k$ for each body, with returns kept in that body and canonical reassociations.

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Data-Availability Statement. The paper and appendix contain rules, proofs and examples.

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A Typing and interpreting the clauses

A.1 Syntax

$$
\begin{aligned}
a, b &::= 0 \mid 1 \mid a + b \mid a \otimes b \\
v &::= * \mid x \mid \mathsf{left}\ v \mid \mathsf{right}\ v \mid \langle v, v \rangle \mid \mathsf{exit}\ v \mid \mathsf{back}\ v \\
p, q &::= * \mid x \mid \langle p, q \rangle \\
e &::= v \mid \textstyle\sum_i \alpha_i e_i \mid \mathsf{let}\ p = \omega\, q\ \mathsf{in}\ e \\
\omega &::= f \mid \{v_i \leftrightarrow e_i\}_{i \in I} \mid \omega\,;\, \omega \mid \omega \oplus \omega \mid \omega \otimes \omega \mid \omega^{\dagger} \mid \mathsf{trace}_c\ \omega \\
t &::= e \mid \omega\, t \\
decl &::= f :: a \leftrightarrow b\,;\ f = \omega \mid \mathsf{round}\ f\ :\ a + c = \{v_i \leftrightarrow e_i\}_{i \in I}
\end{aligned}
$$

Fig. 13. Finite iso definitions. A name f refers to an earlier closed definition. The coefficient α_i is a complex scalar. A signature uses $::$, while the `round` abbreviation uses $:$. The subscript in trace_c identifies the return type.

The surface patterns exit v and back v elaborate recursively to left v and right v. This rewrite preserves payloads and nesting, and is performed before matching or typing by the core rules below. The remaining executable abbreviations are those of Section 2: `Unit`, `unit`, `Qubit` and its basis values, bracketed scalar combinations, flow composition and `on` application. Listings typeset the source token `<->` as $\leftrightarrow$ and pair delimiters `<`, `>` as $\langle$, $\rangle$.

A.2 Statics

Admission, elaboration and evaluation are defined together by induction over the ordered declarations and proper subprograms. In particular, evaluating a local call to fill a table column uses a proper subprogram or an already checked earlier definition. The grammar in Figure 13 describes definitions in an ordered environment Σ. A binding $f = \omega : a \leftrightarrow b$ is introduced after checking ω against earlier bindings. Write Σ ok when these declarations form a well-typed environment. Types are the finite trees of Figure 13. Both 0 and 1 are types, and $a + b$ and $a \otimes b$ are types whenever a and b are. Names may be chosen fresh by renaming. The rules for introducing them are

$$
\frac{}{\emptyset\ \mathsf{ok}} \qquad \frac{\Sigma\ \mathsf{ok} \quad f \notin \mathrm{dom}(\Sigma) \quad \Sigma \vdash \omega : a \leftrightarrow b}{\Sigma, (f = \omega : a \leftrightarrow b)\ \mathsf{ok}}.
$$

The declaration

$$
\mathtt{round}\ f\ :\ a + c = \{v_i \leftrightarrow e_i\}_{i \in I}
$$

introduces the ordinary binding $f = \{v_i \leftrightarrow e_i\}_{i \in I} : (a + c) \leftrightarrow (a + c)$ after the literal clauses have passed the same check. The name denotes the open round. Closing it requires an explicit operation. Writing $\Sigma \vdash d \Rightarrow \Sigma'$ for declaration elaboration, its rule is

$$
\frac{\Sigma\ \mathsf{ok} \quad f \notin \mathrm{dom}(\Sigma) \quad \Sigma \vdash \mathscr{T} : (a + c) \leftrightarrow (a + c)}{\Sigma \vdash \mathtt{round}\ f\ :\ a + c = \mathscr{T} \Rightarrow \Sigma, (f = \mathscr{T} : (a + c) \leftrightarrow (a + c))}.
$$

An ordinary iso signature still permits distinct external types. The binding $f = \omega : a \leftrightarrow b$ represents the surface declaration $f :: a \leftrightarrow b\,;\ f = \omega$; listings may separate these parts by a newline. Write Δ for a finite set of distinct typed value variables. In Δ_1, Δ_2 the two domains must be disjoint. Each judgment below uses this convention, including inside a clause with an empty set of variables. The empty table presents the unique map on 0. Whenever we use a unit input or a nonzero operator line we assume the corresponding external type is inhabited.

Patterns and expressions. The finite basis is defined recursively:

$$\mathscr{B}(0) = \emptyset, \qquad \mathscr{B}(1) = \{*\},$$
$$\mathscr{B}(a+b) = \mathrm{left}\,\mathscr{B}(a) \sqcup \mathrm{right}\,\mathscr{B}(b), \quad \mathscr{B}(a \otimes b) = \mathscr{B}(a) \times \mathscr{B}(b).$$

An environment for Δ assigns each variable a basis value of its type. A variable matches any such value; a constructor matches its own tag, and a pair matches its two components. Write $\Delta \vdash v : a$ when a basis pattern is well typed. The rules are

$$\emptyset \vdash * : 1 \qquad x : a \vdash x : a \qquad \frac{\Delta \vdash v : a}{\Delta \vdash \mathrm{left}\ v : a+b} \qquad \frac{\Delta \vdash v : b}{\Delta \vdash \mathrm{right}\ v : a+b}$$

$$\frac{\Delta_1 \vdash v : a \qquad \Delta_2 \vdash w : b}{\Delta_1, \Delta_2 \vdash \langle v, w \rangle : a \otimes b}.$$

For a context Δ, let $\mathscr{B}(\Delta)$ be its basis assignments and $\mathscr{H}(\Delta) = \mathbb{C}^{\mathscr{B}(\Delta)}$. A product pattern p with $\Delta \vdash p : b$ gives a bijection $\gamma \mapsto p[\gamma]$ from $\mathscr{B}(\Delta)$ to $\mathscr{B}(b)$. For unit it maps the unique empty assignment to $*$. For a variable it is the identity. For a pair, disjoint contexts let us combine the two component assignments uniquely. Induction on these three rules proves the bijection, including empty bases. Matching p is its inverse.

The unit, variable and pair rules also type the product patterns p, q in Figure 13. In the argument position q these rules assemble already bound values. In the result position p they bind fresh variables of the indicated types. A single variable in either position may stand for a value of any type. For expressions we write $\Sigma; \Delta \vdash e : a$ and use

$$\frac{\Delta \vdash v : a}{\Sigma; \Delta \vdash v : a} \qquad \frac{\Sigma; \Delta \vdash e_i : a\ (i \in I) \qquad \alpha_i \in \mathbb{C}}{\Sigma; \Delta \vdash \sum_{i \in I} \alpha_i e_i : a}$$

$$\frac{\Delta_1 \vdash q : a \qquad \Sigma \vdash \omega : a \leftrightarrow b \qquad \Delta_p \vdash p : b \qquad \Sigma; \Delta_2, \Delta_p \vdash e : d}{\Sigma; \Delta_1, \Delta_2 \vdash \mathsf{let}\ p = \omega\, q\ \mathsf{in}\ e : d}.$$

In the last rule the variables in p are fresh and pairwise distinct. A product pattern of type b binds every basis output, by the bijection above. The sum rule permits an empty index set in any stated context Δ; its denotation is the zero linear map from $\mathscr{H}(\Delta)$. Such an expression cannot supply an entire column of a nonempty unitary table. A finite first-order term is a closed expression, or an application of a closed iso to a term of its domain. Writing $\Sigma \vdash t : a$ for a closed term, its rules are

$$\frac{\Sigma; \emptyset \vdash e : a}{\Sigma \vdash e : a} \qquad \frac{\Sigma \vdash t : a \qquad \Sigma \vdash \omega : a \leftrightarrow b}{\Sigma \vdash \omega\, t : b}.$$

This suffices for feeding a superposition of ground values to a program.

For a matrix M with input columns s and output rows t, row s of $M^\dagger$ is the conjugate of column s of M. Consequently its Gram matrix has entries

$$(M^\dagger M)_{ss'} = \sum_{t \in \mathscr{B}(b)} \overline{M_{ts}} M_{ts'}.$$

Each product pairs amplitudes in the same output row. At $s = s'$ it contributes to the squared norm of one column; at $s \neq s'$ it contributes to the inner product between two columns.

Let M have one column for each $s \in \mathscr{B}(a)$, filled with the coefficients of the expression in the uniquely matching clause under its ground substitution. The following rule imposes a finite

extensional check. Its orthogonality premise concerns fully expanded outputs, so it need not infer linear independence from surface syntax.

$$\frac{\Delta_i \vdash v_i : a \quad \Sigma; \Delta_i \vdash e_i : b \quad (i \in I) \qquad \big(v_i[\mathscr{B}(\Delta_i)]\big)_{i\in I} \text{ disjointly covers } \mathscr{B}(a) \qquad |\mathscr{B}(a)| = |\mathscr{B}(b)| \quad M^\dagger M = I}{\Sigma \vdash \{v_i \leftrightarrow e_i\}_{i\in I} : a \leftrightarrow b}$$

Here $v_i[\mathscr{B}(\Delta_i)]$ is the set of all ground instances of pattern v_i. The coverage premise requires these sets to be disjoint and their union to be $\mathscr{B}(a)$. An empty instance set is permitted and contributes no column. This rule is exact over a specified coefficient domain with exact equality and arithmetic, for example rational complex numbers. With mathematical complex parameters it is a well-formedness predicate; exact equality of arbitrary descriptions of complex numbers is not assumed decidable. The finite semantics does not make floating-point tolerance an orthogonality proof.

Reversible combinators. For $M \in \mathbb{C}^{\mathscr{B}(b)\times\mathscr{B}(a)}$ the atom rule is

$$\frac{|\mathscr{B}(a)| = |\mathscr{B}(b)| \qquad M^\dagger M = I}{[M] : a \leftrightarrow b}.$$

Omit the common Σ in the following rules. A bound name has the type at its declaration. For finite types we have the structural isos of Π as abbreviations for the corresponding permutation clauses, including identity, symmetries and distributivity. The reversible rules are

$$\frac{f : a \leftrightarrow b \in \Sigma}{f : a \leftrightarrow b} \qquad \frac{u : a \leftrightarrow b \qquad v : b \leftrightarrow d}{u\,;\, v : a \leftrightarrow d} \qquad \frac{u : a \leftrightarrow b}{u^\dagger : b \leftrightarrow a}$$

$$\frac{u : a \leftrightarrow b \qquad v : d \leftrightarrow f}{u \oplus v : a + d \leftrightarrow b + f} \qquad \frac{u : a \leftrightarrow b \qquad v : d \leftrightarrow f}{u \otimes v : a \otimes d \leftrightarrow b \otimes f} \qquad \frac{\beta : a + c \leftrightarrow b + c}{\mathrm{trace}_c\, \beta : a \leftrightarrow b}.$$

A.3 Dynamics

Evaluating an expression. Let $\gamma \in \mathscr{B}(\Delta)$ assign basis values to the variables in a linear context Δ, and write $\gamma \vdash e \Downarrow \psi$ when evaluation returns the finite output vector ψ. For a value pattern v, $v[\gamma]$ is its ground instance. Values and sums evaluate by

$$\frac{}{\gamma \vdash v \Downarrow |v[\gamma]\rangle} \qquad \frac{\gamma \vdash e_i \Downarrow \psi_i \ (i \in I)}{\gamma \vdash \sum_i \alpha_i e_i \Downarrow \sum_i \alpha_i \psi_i}.$$

For let $p = \omega q$ in e, split the input assignment as $\gamma_1 \uplus \gamma_2$ according to the typing rule: q uses precisely γ_1, and e retains γ_2. Write $U|q[\gamma_1]\rangle = \sum_t a_t|t\rangle$, where $\omega \Downarrow U$. Matching the product pattern p with t gives the fresh assignment γ_t. The application rule is

$$\frac{\omega \Downarrow U \quad \gamma_2 \uplus \gamma_t \vdash e \Downarrow \psi_t \quad (t \in \mathscr{B}(b))}{\gamma_1 \uplus \gamma_2 \vdash \text{let } p = \omega q \text{ in } e \Downarrow \sum_t a_t \psi_t}.$$

Here b is the output type of ω. Its admitted matrix supplies the coefficients a_t, and total matching of p supplies each γ_t. The same binding is used independently in the alternatives of a sum; it is split only where the linear typing rule splits it. For closed terms, $t \Downarrow \psi$ uses the empty variable assignment, with

$$\frac{\varnothing \vdash e \Downarrow \psi}{e \Downarrow \psi} \qquad \frac{\omega \Downarrow U \qquad t \Downarrow \psi}{\omega t \Downarrow U\psi}.$$

The fixed iso environment is omitted from these evaluation judgments. Named matrices are computed from earlier definitions; the reversible combinators act by their matrix operations, and a trace uses Proposition B.2. Thus expression evaluation is finite even when its matrix contains feedback

with infinitely many nonzero exits. Induction on an expression's typing derivation shows that evaluation gives its linear interpretation: the value rule substitutes basis values, the sum rule is linearity, and the application rule composes the admitted iso with the continuation's linear map. These rules do not select a measurement outcome or normalize individual amplitude components. More explicitly, the local application's column at (γ_1, γ_2) is

$$E_{\mathsf{let}}(s; \gamma_1, \gamma_2) = \sum_{t \in \mathscr{B}(b)} U_{t,q[\gamma_1]}\, E_e(s; \gamma_2, \gamma_t).$$

Here $s \in \mathscr{B}(d)$ indexes the continuation output, and $t \in \mathscr{B}(b)$ indexes the intermediate output of the local iso. This is ordinary matrix multiplication: reassemble q's variables, apply U while carrying γ_2, match its output to the fresh product pattern, and apply the continuation matrix. The two pattern maps are the bijections just proved. Thus the rule acts linearly on arbitrary superpositions of assignments, including entangled ones. It neither duplicates an assignment nor measures the local result.

Elaboration and matrix evaluation. The matrix programs have grammar

$$\kappa ::= [M] \mid \kappa; \kappa \mid \kappa \oplus \kappa \mid \kappa \otimes \kappa \mid \kappa^\dagger \mid \mathsf{trace}_c\, \kappa.$$

The combinators κ of Section 2 have ground checked tables as atoms. Write $\mathscr{E}_\Sigma(\omega)$ for elaboration. If $f = \omega_f$ is a declaration in Σ, let Σ_f be its earlier prefix, the environment in which ω_f was checked. With M the evaluated columns of a table $\mathscr{T}$, the definition is

$$\begin{aligned}
\mathscr{E}_\Sigma(\mathscr{T}) &= [M], & \mathscr{E}_\Sigma(f) &= \mathscr{E}_{\Sigma_f}(\omega_f),\\
\mathscr{E}_\Sigma(\omega; \nu) &= \mathscr{E}_\Sigma(\omega); \mathscr{E}_\Sigma(\nu), & \mathscr{E}_\Sigma(\omega \oplus \nu) &= \mathscr{E}_\Sigma(\omega) \oplus \mathscr{E}_\Sigma(\nu),\\
\mathscr{E}_\Sigma(\omega \otimes \nu) &= \mathscr{E}_\Sigma(\omega) \otimes \mathscr{E}_\Sigma(\nu), & \mathscr{E}_\Sigma(\omega^\dagger) &= \mathscr{E}_\Sigma(\omega)^\dagger,\\
\mathscr{E}_\Sigma(\mathsf{trace}_c\, \omega) &= \mathsf{trace}_c\, \mathscr{E}_\Sigma(\omega).
\end{aligned}$$

These equations recurse into proper subprograms and, on lookup, earlier definitions. Computing a table uses this same evaluation on its local iso subprograms. This is a well-founded mutual definition with expression evaluation. Each clause of the definition preserves its source and target types. The declaration rule for `round` binds this open combinator and does not insert a trace.

If $\kappa \Downarrow M$ and $\lambda \Downarrow N$, the ordinary matrix rules, used with their typed premises, are

$$\begin{gathered}
[M] \Downarrow M, \qquad (\kappa; \lambda) \Downarrow NM,\\
(\kappa \oplus \lambda) \Downarrow M \oplus N, \quad (\kappa \otimes \lambda) \Downarrow M \otimes N, \quad \kappa^\dagger \Downarrow M^\dagger.
\end{gathered}$$

The feedback rule completes them:

$$\frac{\kappa \Downarrow U \quad U = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) \quad F : X \to Y \quad (I_Y - D)F = C}{\mathsf{trace}_c\, \kappa \Downarrow A + BF}.$$

The premise solves a finite linear system. It neither inverts a possibly singular $I - D$ nor requires an execution to reach a last exit. Theorem B.1 gives a solution and proves that every solution gives the same output. Proposition B.2 computes that output by eliminating one return pattern at a time. Thus this relational evaluation has a result independent of the chosen wire solution, and exact elimination implements it. Write $\Sigma \vdash \omega \Downarrow M$ when its elaboration evaluates to M. Closed application then uses

$$\frac{\Sigma \vdash \omega \Downarrow M \qquad t \Downarrow \psi}{\omega t \Downarrow M\psi}.$$

For a well-typed expression and a basis assignment, induction on its syntax gives a unique output vector. In the local-binding case, the argument and result patterns give inverse assembly/matching

bijections, and the finite coefficient sum is the matrix product displayed above. Expanding each clause therefore preserves all its columns. Induction through the combinators preserves evaluation and the linear denotation; the trace case uses the feedback theorem. Linearity then gives the same result on arbitrary superpositions, including assignments entangled with a retained factor.

The executable listings use a finite clause frontend and a flow frontend for the surrounding combinators. Their flow spellings are ; for sequence, + for direct sum and ∗ for tensor. Adjoint is available mathematically by transposing and conjugating the ground table. Unit and nested local patterns, the empty type and the empty table belong to the mathematical grammar; the listings use variable or flat-tuple local bindings and inhabited finite types. Native numerical evaluation uses floating-point coefficients and may refuse a coupled near-unit pivot; it is not the exact field-arithmetic procedure of Proposition B.2.

A.4 Denotation

A checked table denotes the linear extension of its columns. Names, sequence, dagger, direct sum and tensor denote lookup, composition in execution order, adjoint, direct sum and tensor; trace denotes the exit sum (3.2). The evaluation identities above give these same maps, with the trace case supplied by Theorem B.1.

Control of an endo-iso u is $I \oplus u$ under the distributive iso for $(1+1)\otimes a$. Thus controlling a global phase can change a relative phase. The types above deliberately do not provide this operation on an arbitrary channel.

A.5 Effect extensions

Syntax. Write u for a reversible iso, v for an isometry program and h for a channel program. The two upper grammars are

$$v ::= \mathsf{arr}\, u \mid \mathsf{lift}\, u \mid v\,;\, v \mid v \oplus v \mid v \otimes v \mid \mathsf{count}_K\, u,$$
$$h ::= \mathsf{arr}\, v \mid \mathsf{hide}\, v \mid h\,;\, h \mid h \otimes h \mid \mathsf{while}_c\, u.$$

Statics. The type of the intermediate result must match the type expected by the next program. At the isometry layer this gives the following rules for sequence, tensor and direct sum.

$$\frac{v : a \rightarrowtail b \qquad w : b \rightarrowtail d}{v\,;w : a \rightarrowtail d} \qquad \frac{v : a \rightarrowtail b \qquad w : c \rightarrowtail d}{v \otimes w : a \otimes c \rightarrowtail b \otimes d} \qquad \frac{v : a \rightarrowtail b \qquad w : c \rightarrowtail d}{v \oplus w : a + c \rightarrowtail b + d}.$$

Channels also compose in sequence and by tensor product.

$$\frac{h : a \rightsquigarrow b \qquad k : b \rightsquigarrow d}{h\,;k : a \rightsquigarrow d} \qquad \frac{h : a \rightsquigarrow b \qquad k : c \rightsquigarrow d}{h \otimes k : a \otimes c \rightsquigarrow b \otimes d}.$$

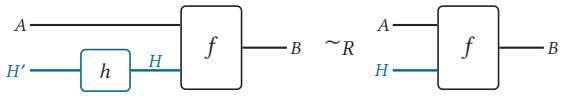
These operations have their usual linear meanings, with sequence written in execution order. An identity on each layer is the embedded identity iso. The remaining rules change the kind of map or close a round.

$$\frac{u : a \leftrightarrow b}{\mathsf{arr}\, u : a \rightarrowtail b} \qquad \frac{u : a + d \leftrightarrow b}{\mathsf{lift}\, u : a \rightarrowtail b} \qquad \frac{\beta : a + c \leftrightarrow b + c \qquad K \geq 1}{\mathsf{count}_K\, \beta : a \rightarrowtail [K] \otimes b}$$

$$\frac{v : a \rightarrowtail b}{\mathsf{arr}\, v : a \rightsquigarrow b} \qquad \frac{v : a \rightarrowtail b \otimes g \qquad |\mathscr{B}(g)| > 0}{\mathsf{hide}\, v : a \rightsquigarrow b} \qquad \frac{\beta : a + c \leftrightarrow b + c}{\mathsf{while}_c\, \beta : a \rightsquigarrow b}.$$

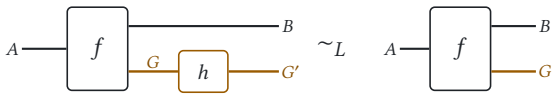
Denotation. An $R[\mathbf{U}]$ representative $U : X \oplus H \to X'$ exposes $u \mapsto U(u, 0)$, identifying changes confined to the unused summand. An $L[\mathbf{I}]$ representative $V : X \to X' \otimes G$ exposes $\rho \mapsto \mathrm{tr}_G(V \rho V^\dagger)$, identifying isometric relabelings of the hidden factor and the equivalence relation they generate [11, Prop. 7, Thm. 22].

(a) $R[\mathscr{C}]$: a heap enters beside the input, and only access to it counts



normal form: $A \to A \oplus H$, then f;
$R[\text{Unitary}] \simeq \text{Isometry}$

(b) $L[\mathscr{C}]$: garbage leaves beside the output, and its contents do not count



normal form: f, then $B \otimes G \to B$;
$L[R[\text{Unitary}]] \simeq \text{CPTP}$

Fig. 14. Allocation exposes an input summand of a larger reversible map. Hiding exposes one output factor of an isometry. Preprocessing the unused summand or isometrically relabeling the hidden factor preserves the exposed map.

The allocation permutation uses heap $[K-1] \otimes a$. The basis copier uses a heap with $|\mathscr{B}(b)|^2 - |\mathscr{B}(b)|$ basis values, completing the diagonal injection to a permutation. An embedded unitary is the same map viewed as an isometry. An embedded isometry V is the channel $\mathscr{E}_V$, distinct from the syntactic elaboration $\mathscr{E}_\Sigma$. For lift and hide the denotations are restriction and partial trace as in Section 4. For count_K, tensor $I_{[K]}$ with β, distribute the tensor over the two tags and append $I_{[K] \otimes b} \oplus (S_K \otimes I_c)$ in the resulting order. Trace the summand $[K] \otimes c$, then precompose with $u \mapsto |0\rangle \otimes u$. This allocation is a lift of a permutation that places the exposed basis at counter value zero. The four blocks of the enlarged iso are $I \otimes A, I \otimes B, S_K \otimes C, S_K \otimes D$. Its kth exit is $S_K^k \otimes x_k$, proving (4.1). To hide the counter in the displayed syntax, compose with $\mathrm{arr}\, \sigma_{[K],b}$ before hide. Native `count K beta` returns data before counter, so its following `discard` hides the counter. Appending `swap` instead gives our displayed order. In the flow frontend, `discard` is the derived first projection, $I_a \otimes \mathsf{discard}_b; \mathsf{unit}_\otimes$, rather than the whole-register operation $\mathsf{discard}_b : b \rightsquigarrow 1$.

For while, allocate a fresh bit at 0 and apply the ordinary iso

$$\omega_{\mathrm{tag}} = \{ \begin{array}{l} \langle \mathsf{exit}\, x, 0\rangle \leftrightarrow \langle \mathsf{exit}\, x, 0\rangle, \\ \langle \mathsf{exit}\, x, 1\rangle \leftrightarrow \langle \mathsf{exit}\, x, 1\rangle, \\ \langle \mathsf{back}\, y, 0\rangle \leftrightarrow \langle \mathsf{back}\, y, 1\rangle, \\ \langle \mathsf{back}\, y, 1\rangle \leftrightarrow \langle \mathsf{back}\, y, 0\rangle \quad \} \end{array} : (b + c) \otimes \texttt{Qubit} \leftrightarrow (b + c) \otimes \texttt{Qubit}.$$

The patterns partition the basis and merely permute its values, so the column test admits this iso. Its restriction to bit input 0 gives the isometry T. For output components $v \in [\![b]\!]$ and $y \in [\![c]\!]$, its linear action is

$$T(v, y) = (\iota_{\mathrm{ext}} v) \otimes |0\rangle + (\iota_{\mathrm{ret}} y) \otimes |1\rangle.$$

For the empty type, $\mathsf{discard}_0$ is obtained by hiding the unit factor of the unique isometry $0 \rightarrowtail 1 \otimes 1$, constructed by lift from $0 + 1 \leftrightarrow 1 \otimes 1$. Its action is the unique map on the zero operator space. Set $\mathsf{measure}_0$ to the embedded identity on 0. These supply the empty cases omitted from the inhabited-type formulas in the body.

Applying T after U and tracing out the copied bit gives the diagonal output blocks. Feedback now uses the classical sum. This construction works when the payload types differ. The notation while names this classical feedback operation on the translated body.

When $a = b = c$, let factor $: a + a \leftrightarrow a \otimes \texttt{Qubit}$ and dist $=$ factor†, and put $\hat{\beta} =$ dist; β; factor. Writing $E(\omega) = \mathsf{arr}(\mathsf{arr}\,\omega)$, the channel in Figure 8 is

$$E(\mathsf{factor}); E(\hat{\beta}); (I_a \otimes \mathsf{measure}_{\texttt{Qubit}}); E(\mathsf{dist}).$$

The first factor; dist cancel, leaving the original body followed by exposure, measurement and restoration of its tag.

Soundness. Induct on the finite syntax in a well-founded order that puts prior definitions before uses. The unique match and the substitution clause give a column at each closed input. The admission premise makes these columns an orthonormal basis. Sequential, sum, tensor and dagger operations preserve the stipulated map classes. Theorem B.1 proves the trace case. Restriction of a unitary is an isometry, and a unitary after counter allocation is an isometry. Embedding an isometry and taking a partial trace after an isometry both give completely positive trace-preserving maps. The positive sum in (4.2) is completely positive, since each finite partial sum is and the completely positive cone in finite dimension is closed; its trace preservation follows from $\sum_k x_k^\dagger x_k = I$. Thus every well-typed iso, isometry program and channel program denotes the indicated kind of map. A closed term $t : a$ instead denotes its evaluated vector in $\mathbb{C}^{\mathscr{B}(a)}$; its typing alone does not require that vector to have norm one. Conversely a given finite unitary V has as a definition the table $s \leftrightarrow \sum_t V_{ts} t$. Each column satisfies the typing rule since $V^\dagger V = I$, including the empty case.

B Convergence, probability and elimination

For a finite permutation, an external input follows a single path. It cannot revisit an internal basis value: unique predecessors would identify its original external input with an internal value. It therefore exits within $|\mathscr{B}(c)| + 1$ calls, while wholly internal cycles remain unreachable. The two external paths in Figure 15 combine to a swap, as in reversible flowchart semantics [10, Prop. 3.10][14, Prop. 5.3].

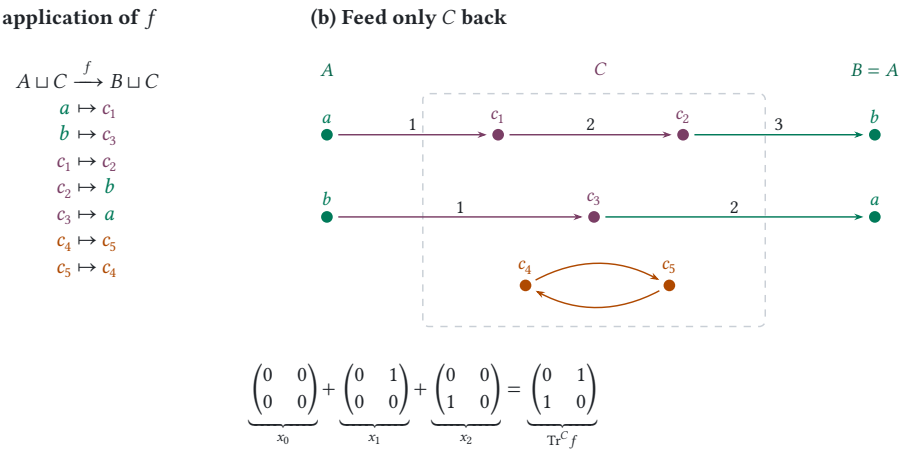
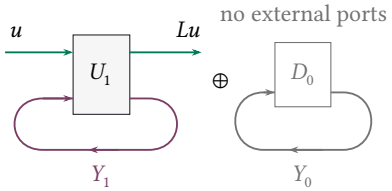


Fig. 15. A reversible loop with two external paths and an internal cycle. The two nonzero exit maps have disjoint domains and their sum is a swap.

Theorem B.1 (Finite feedback). *For a finite unitary $U : X \oplus Y \to X' \oplus Y$, the exits are absolutely summable in operator norm. Their sum $L : X \to X'$ is unitary, and $\sum_k x_k^\dagger x_k = I_X$. The wire equation $y = Cu + Dy$ has a solution for every u. Every solution gives the same output $Au + By = Lu$.*



$$\|u\|^2 = \|Au\|^2 + \|Cu\|^2,$$
$$\|Cu\|^2 = \|BCu\|^2 + \|DCu\|^2,$$
$$\|DCu\|^2 = \|BDCu\|^2 + \|D^2Cu\|^2.$$

Fig. 16. An internal basis change separates U_1 from the unreachable D_0. The three norm identities telescope because each return term supplies the next call.

Proof of Theorem B.1. From $U^\dagger U = I$ and $UU^\dagger = I$ we have

$$A^\dagger A + C^\dagger C = I, \qquad B^\dagger B + D^\dagger D = I, \qquad CC^\dagger + DD^\dagger = I.$$

The middle identity gives $\|Dy\|^2 + \|By\|^2 = \|y\|^2$. With $Y_0 = \bigcap_{j\ge 0} \ker BD^j$, it follows that $DY_0 \subseteq Y_0$ and $D|_{Y_0}$ is isometric. In finite dimension it maps Y_0 onto Y_0. Thus U maps $0_X \oplus Y_0$ onto $0_{X'} \oplus Y_0$. Unitarity maps their orthogonal complements $X \oplus Y_1$ and $X' \oplus Y_1$ onto one another. Hence $C(X) \subseteq Y_1$, $D(Y_1) \subseteq Y_1$ for $Y_1 = Y_0^\perp$, and B vanishes on Y_0.

Write $D_1 = D|_{Y_1}$ and $B_1 = B|_{Y_1}$. Put $m = \dim Y$. By Cayley–Hamilton a return vector is invisible exactly when its first m emissions vanish. If $Y_1 = 0$, all delayed exits vanish. Otherwise $m > 0$. For a unit $y \in Y_1$, conservation through m calls gives

$$1 - \|D_1^m y\|^2 = \sum_{j=0}^{m-1} \|BD^j y\|^2 > 0.$$

Equality would make every displayed emission zero and hence put y in Y_0. On the compact unit sphere of Y_1, the continuous function $y \mapsto \|D_1^m y\|$ therefore has a maximum $q < 1$. For $k = pm + r$, $0 \le r < m$, norm nonincrease on the remaining r calls gives $\|D_1^k\| \le \|D_1^m\|^p \|D_1^r\| \le q^p$. This geometric bound also makes every polynomially weighted exit series converge. Therefore $\sum \|x_k\|$ is finite and $F = (I - D_1)^{-1}C = \sum_j D_1^j C$ solves the wire equation. Every other solution differs by a vector in Y_0 fixed by D, which B kills. The norm identity for $U(u, Fu) = (Lu, Fu)$ proves that L is an isometry. Since $\dim X = \dim X'$ it is onto.

To account for the probability, let $T_j = C^\dagger D^{\dagger j} D^j C$. On a unit input u, $\langle u, T_j u\rangle$ is the probability still inside after call $j+1$. The first block identity gives $x_0^\dagger x_0 = I - T_0$. On later calls the weight that leaves is exactly the drop in the weight still inside, since

$$x_{j+1}^\dagger x_{j+1} = C^\dagger D^{\dagger j} \underbrace{B^\dagger B}_{I - D^\dagger D} D^j C = T_j - T_{j+1}.$$

Adding through exit k gives $\sum_{j=0}^{k} x_j^\dagger x_j = I_X - T_k$. The residual term tends to zero by the decay just proved, giving total exit probability one even for a body with infinitely many possible exits. □

For an internal state w, write $\mathcal{O}w = (BD^j w)_{j\ge 0}$ for the sequence of amplitudes it can still emit. Let $\mathcal{S}(z_0, z_1, \ldots) = (z_1, z_2, \ldots)$ be the left shift. Then $\mathcal{O}D_1 = \mathcal{S}\mathcal{O}$ on Y_1 says that advancing a live state drops its first emission from the sequence. The squared norms of the entries of $\mathcal{O}w$ sum to $\|w\|^2$ for $w \in Y_1$, by the same conservation argument as in the proof. Figure 17 illustrates this with $B = -4/5$ and $D = 3/5$ on the live direction, and with an independent dead direction.

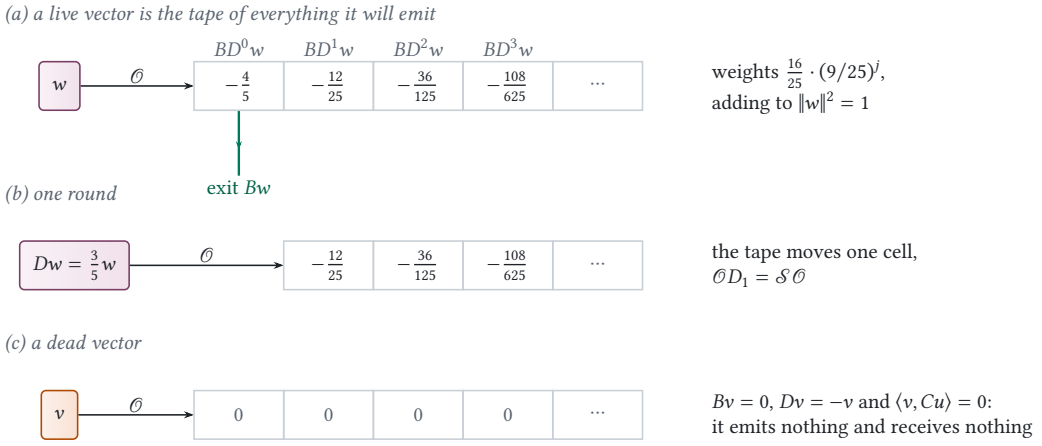


Fig. 17. The future emissions of a live state form its observation tape. Advancing that state shifts the tape by one position. A dead state has the zero tape. Here $Dv = -v$ is dead but not fixed; only the fixed subspace $\ker(I - D)$ obstructs the inverse at phase one.

B.1 Eliminating a pattern

Proposition B.2 (Elimination). *Given the expanded table, eliminating its returning basis values in any order produces a closed iso denoting L in exactly $|\mathscr{B}(c)|$ steps. If the original coefficients lie in a subfield of $\mathbb{C}$, the result has coefficients in that subfield.*

Proof of Proposition B.2. Place the selected return pattern in $Q = \mathbb{C}q$ and put $Y_{\text{rest}} = Y \ominus Q$, the orthogonal complement of Q in Y. The remaining input and output are $E_{\text{in}} = X \oplus Y_{\text{rest}}$ and $E_{\text{out}} = X' \oplus Y_{\text{rest}}$. Grouping those remaining ports together gives body blocks

$$\begin{pmatrix} M & b_q \\ c_q & d_q \end{pmatrix}, \qquad M : E_{\text{in}} \to E_{\text{out}}, \quad b_q : Q \to E_{\text{out}}, \quad c_q : E_{\text{in}} \to Q, \quad d_q \in \mathbb{C}.$$

Here $d_q = D_{qq}$ is the selected self-entry. For $w = (u, y_{\text{rest}})$, the row c_q gives $c_q w = (Cu)_q + \sum_{r \neq q} D_{qr} y_r$ from the main-text coordinate equation, while b_q collects the coefficients of y_q in the other outputs. Closing Q asks for a scalar $\lambda = c_q w + d_q \lambda$ and returns $Mw + b_q \lambda$. When $d_q \neq 1$, solving gives $\lambda = c_q w/(1 - d_q)$ and output $(M + b_q c_q/(1 - d_q))w$. When $d_q = 1$, its self-entry has norm one, so every other entry in both its row and column is zero. Hence $b_q = c_q = 0$, any λ solves the wire equation, and the output is Mw. These cases are exhaustive. In particular $d_q = -1$ or $d_q = i$ uses the first branch, with $b_q = c_q = 0$; no divergent geometric series is invoked. Theorem B.1 applied with these remaining ports external proves unitarity of this intermediate table in either case.

Figures 3 and 4 apply this calculation to the scalar rotation and the two return patterns of the flip.

After one elimination, solving the remaining feedback equations gives exactly the output of solving all the original equations. Indeed a simultaneous solution restricts to one of the smaller system, and a solution of the smaller system can be extended by a solution for q at those incident values. The external value is independent of these choices by Theorem B.1. Induct on the number of return patterns. Each step removes one and uses only field operations with nonzero denominators. For a phase on the return wire apply the same argument to the modified unitary. No exchange in the order of a conditionally convergent series of individual walks is needed. □

Closed sums for the running rotation. For `guardRound`, both port spaces are $\mathbb{C}$, so the blocks are scalars: $A = D = t = 3/5$, $C = s = 4/5$ and $B = -s$. The general exit formula becomes $x_0 = t$ and $x_{k+1} = (-s)t^k s = -s^2 t^k$. To compare adding these amplitudes with adding their squared magnitudes, follow the first n calls on unit input. Write r_n for the amplitude still returning, ℓ_n for the sum emitted so far, and p_n for the weight of those exits when the guard is measured. The geometric sums give

$$r_n = st^{n-1},$$
$$\ell_n = t - s^2 \sum_{j=0}^{n-2} t^j = -1 + (1+t)t^{n-1},$$
$$p_n = t^2 + s^4 \sum_{j=0}^{n-2} t^{2j} = 1 - s^2 t^{2(n-1)}.$$

The sums are empty when $n = 1$. Table 1 follows the resulting emissions, residuals and sums. The squared length $|\ell_n|^2$ can fall and rise as later emissions interfere. The measured halting weight p_n only increases, because each call adds a nonnegative probability.

A table and an unknown gate. There is also no uniform approximation guarantee for an unknown body given only as a gate. For a scalar rotation with cosine t the trace is $t - (1-t^2)/(1-t) = -1$ for $-1 < t < 1$, but at the identity rotation it is 1. Put a bypass beside this scalar and the jump is a relative phase detectable by a Hadamard. A fixed finite circuit built by continuously composing the supplied body has a continuous response; eliminating a given clause table can branch on an exact equality [9, Remark B.1].

C Records, measurement and phase

The counter and measured-loop constructions appear in Figures 6, 7 and 8 in the main text. For a common payload $H = X = X' = Y$, the guard-first transport is $F_g = \mathrm{factor}; \sigma_{H,\mathtt{Qubit}} : H + H \leftrightarrow \mathtt{Qubit} \otimes H$. It gives $\widehat{U}_g = [\![F_g]\!] U [\![F_g]\!]^{-1}$. For a counter, $F_{K,g} = d_K^{-1}; (I_{[K]} \otimes F_g)$ puts counter, guard and data in that order, with inverse transport at the output. These are the equal-payload forms of the general constructions in Section 4.

Use the Hilbert space $H_{\mathrm{rec}} = \ell^2(\mathbb{N})$ for an exact record, in the order counter then data. The squared norm of Ru is $\sum_k \|x_k u\|^2 = \|u\|^2$. For the isometric inclusion $J_K : \mathbb{C}^K \to H_{\mathrm{rec}}$ onto the first K labels,

$$\|(J_K \otimes I)R_K - R\| \;\le\; \sum_{k \ge K} \big\| |k \bmod K\rangle - |k\rangle \big\| \, \|x_k\| \;\le\; 2 \sum_{k \ge K} \|x_k\|.$$

For $k \ge K$, the two distinct orthogonal count labels have distance $\sqrt{2} \le 2$. The bound therefore tends to zero. This argument only compares the operators produced by finite bodies. The map that folds all square-summable sequences into residue classes would be unbounded, as would the map that sums their entries at phase one.

For an input of the form $|u\rangle\langle v|$, tracing the exact record over its orthonormal basis leaves $\sum_k x_k |u\rangle\langle v| x_k^\dagger$. Such operators span the input matrix algebra, giving (4.2). For a finite counter put $G_j = \sum_{k \equiv j \bmod K} x_k$. Discarding that counter instead gives $\sum_{j<K} G_j \rho G_j^\dagger$. As a result, unequal counts congruent modulo K retain cross terms. For the Fourier forms, the coefficient sum is absolutely convergent, and

$$G_j = \frac{1}{K} \sum_{z^K = 1} \bar{z}^j \Theta(z), \qquad x_k = \frac{1}{2\pi} \int_0^{2\pi} e^{-ik\varphi} \Theta(e^{i\varphi}) \, d\varphi.$$

Expanding $\Theta(z)\rho\Theta(z)^\dagger$ and integrating term by term removes all unequal indices, proving (6.1).

Definition 4.1 in Section 4 collects the resulting maps on the reference example. For the classical trace, let $\theta(\rho,\sigma) = \mathrm{diag}(\rho,\sigma)$ insert the two operator blocks and let φ extract the diagonal output blocks. Then $\Phi_U = \varphi \circ \mathscr{E}_U \circ \theta$ is the branching map of Section 4. Its trace on completely positive trace-nonincreasing maps on finite C^*-algebras is the unrolling $\sum_k x_k \rho x_k^\dagger$ [3, Prop. 3.2.17].

Relation to monitored first returns. The published test-first loop measures before applying U [21, § 3]. To obtain our body-first convention when $X' = X$, apply U once before entering it. Let $\iota_X u = (u, 0)$ and $P = \iota_X \iota_X^\dagger$. For $n \geq 1$, the exit map on call n is $\iota_X^\dagger U((I-P)U)^{n-1}\iota_X = x_{n-1} : X \to X$. Thus the monitored-return count includes the initial call and is one more than our number of returns [6, § 2].

D Agreement and replacement

Proof of Theorem 5.1. Assume $\mathscr{M} = \mathscr{E}_L$. For each unit input u, test the output on any w orthogonal to Lu. The sum of the nonnegative numbers $|\langle w, x_k u\rangle|^2$ is zero. Thus $x_k u$ is parallel to Lu for each k and all u. Put $V = L^\dagger x_k$. Since Vu lies along u, testing a basis vector e_i gives $Ve_i = \lambda_i e_i$. To see why the scalar cannot depend on i, test two distinct basis vectors together:

$$V(e_i + e_j) = \underbrace{\lambda_i e_i + \lambda_j e_j}_{\text{by linearity}} = \underbrace{\lambda_{ij}(e_i + e_j)}_{\text{parallel to the input}} .$$

Comparing the two coordinates gives $\lambda_i = \lambda_{ij} = \lambda_j$. Hence $V = b_k I$ and $x_k = b_k L$, including in dimension one where there is only one basis coefficient to begin with. By absolute convergence, $\sum_k b_k = 1$ since $\sum_k x_k = L \neq 0$. By $\sum_k x_k^\dagger x_k = I$, $\sum_k |b_k|^2 = 1$. The unit vector $\xi = \sum_k b_k |k\rangle$ therefore satisfies $Ru = \xi \otimes Lu$. Conversely that factorization implies $\mathscr{M} = \mathscr{E}_L$ after tracing out the count; testing each label shows $x_k = \langle k|\xi\rangle L$.

These conditions imply that $x_0, \ldots, x_m$ span at most a line. If $m > 0$, Cayley–Hamilton writes D^m as a linear combination of $I, D, \ldots, D^{m-1}$. Since $x_{k+1} = BD^kC$, multiplication by higher powers shows that every exit after x_m belongs to the span of $x_1, \ldots, x_m$. If $m = 0$, all exits after x_0 vanish. Thus the span of the first $m+1$ exits is the span of all the exits. It cannot be zero because it contains their unitary sum on the nonzero input. If it is a line, the same sum L spans it, so $x_k = b_k L$ for each k. Completeness of the exits again gives the unit norm of ξ. This proves all implications. □

Channel contexts. Fix the hole type $a \rightsquigarrow b$ and abbreviate it by $\mathscr{H}$ to the left of a context judgment. The single hole, sequential and tensor constructors in Section 5 have rules

$$\mathscr{H} \vdash [-] : a \rightsquigarrow b \qquad \frac{h : e \rightsquigarrow c \qquad \mathscr{H} \vdash \mathscr{C} : c \rightsquigarrow d}{\mathscr{H} \vdash h;\mathscr{C} : e \rightsquigarrow d} \qquad \frac{\mathscr{H} \vdash \mathscr{C} : c \rightsquigarrow d \qquad h : d \rightsquigarrow e}{\mathscr{H} \vdash \mathscr{C};h : c \rightsquigarrow e}$$

$$\frac{h : e \rightsquigarrow f \qquad \mathscr{H} \vdash \mathscr{C} : c \rightsquigarrow d}{\mathscr{H} \vdash h \otimes \mathscr{C} : e \otimes c \rightsquigarrow f \otimes d} \qquad \frac{\mathscr{H} \vdash \mathscr{C} : c \rightsquigarrow d \qquad h : e \rightsquigarrow f}{\mathscr{H} \vdash \mathscr{C} \otimes h : c \otimes e \rightsquigarrow d \otimes f}.$$

For a finite instrument $(\Psi_j : c \to c_j)_{j \in J}$, meaning $\Psi_j : \mathrm{End}([\![c]\!]) \to \mathrm{End}([\![c_j]\!])$, the maps are completely positive and their output traces sum to the input trace. Its case rule is

$$\frac{i \in J \quad \mathscr{H} \vdash \mathscr{C} : c_i \rightsquigarrow d \quad h_j : c_j \rightsquigarrow d \ (j \neq i)}{\mathscr{H} \vdash \mathsf{case}\big(\Psi_i;\mathscr{C},(\Psi_j;h_j)_{j\neq i}\big) : c \rightsquigarrow d}.$$

Plugging h into the hole interprets sequencing by channel composition and tensor by the tensor product of channels. The case denotes $[\![\mathscr{C}[h]]\!] \circ \Psi_i + \sum_{j\neq i} [\![h_j]\!] \circ \Psi_j$. Thus exactly one syntactic occurrence depends on the hole, and each branch continuation has the channel type stated by the

rules. These contexts can use a hole on one instrument outcome and omit it on the others. They do not contain iso-level operations on that hole.

Proof of Corollary 5.2. If $X = 0$ then $X' = 0$ as well, and there is just one map on the zero operator space. Now suppose $X \neq 0$. On agreement the finite record has the form $R_K u = \xi_K \otimes Lu$, where $\xi_K = \sum_k b_k |k \bmod K\rangle$. Since R_K is an isometry and an input unit vector exists, $\|\xi_K\| = 1$, even if components with equal labels interfere. Discarding it gives $\mathscr{E}_L$. For the exact record the equality after discard needs no agreement condition. Channel equality on data entails equality after tensoring with identity on any finite reference, since elementary tensor operators span the joint matrix algebra. Composition with fixed channels, discarding and applying a fixed instrument on either program therefore preserve equality. For a measurement-conditioned continuation apply the same argument separately to each unnormalized outcome and then add if the outcome is forgotten. This proves the claim by induction over a context built with these constructors. Conversely, if the channels differ, some density operator distinguishes them since positive trace-one inputs span the matrix algebra over $\mathbb{C}$. The two resulting density operators are distinguished by a spectral projection of their Hermitian difference. This test is at the output of the empty context; finite exact isos and tag measurement can implement a measurement in the chosen eigenbasis.

For delayed discard, use the following separate semantic continuation calculus. Its ordinary data types are finite, while its opaque history names may carry countable semantic record spaces. Write $\Gamma \vdash K : a \Rightarrow b$ for a finite continuation that starts with ordinary data of type a and the fresh opaque history names in Γ, and returns data of type b after consuming all those names. Each name carries its designated record space. Each branch Ψ_j of a finite instrument sends operators on $[\![a]\!]$ to operators on $[\![a_j]\!]$ by a completely positive map. The traces of these outputs sum to one on each density input. An ordinary channel is its one-outcome case. For a binder, let $R : [\![a]\!] \to H \otimes [\![d]\!]$ be a recording isometry. Its history name g has space H. The caller rules are

$$\emptyset \vdash \mathsf{return} : a \Rightarrow a \qquad \frac{\Gamma \vdash K_j : a_j \Rightarrow b \quad (j \in J)}{\Gamma \vdash \mathsf{case}\,(\Psi_j; K_j)_{j\in J} : a \Rightarrow b}$$

$$\frac{\Gamma \vdash K : a \Rightarrow b \quad h \notin \Gamma}{\Gamma, h \vdash \mathsf{drop}\ h; K : a \Rightarrow b} \qquad \frac{\Gamma, g \vdash K : d \Rightarrow b \quad g \notin \Gamma}{\Gamma \vdash \mathsf{record}\ g = R\ \mathsf{in}\ K : a \Rightarrow b}.$$

Bound names are chosen fresh throughout the finite derivation by renaming. For an ordering of the live names, let H_Γ be their tensor product and I_Γ its identity. The semantics maps trace-class operators on $H_\Gamma \otimes [\![a]\!]$ to operators on $[\![b]\!]$. With canonical reassociations and permutations of named factors understood, its equations are

$$\begin{aligned}
[\![\mathsf{return}]\!] &= \mathrm{id}, \\
[\![\mathsf{case}\,(\Psi_j; K_j)_{j\in J}]\!] &= \sum_{j\in J} [\![K_j]\!] \circ (\mathscr{E}_{I_\Gamma} \otimes \Psi_j), \\
[\![\mathsf{drop}\ h; K]\!] &= [\![K]\!] \circ \mathrm{tr}_h, \\
[\![\mathsf{record}\ g = R\ \mathsf{in}\ K]\!] &= [\![K]\!] \circ \mathscr{E}_{I_\Gamma \otimes R}.
\end{aligned}$$

The instrument selects a continuation by its classical outcome and passes all live names to that continuation. A binder adds its new record, and a drop traces out just the named factor. Only the empty history context can return, so each returning branch drops every live name exactly once. For K_h starting with a distinguished h, fresh for every internal record binder, delete its drops to obtain K, leaving all other binders and drops in place. The rules show that K is well typed with h

removed. On each returning branch and for any joint trace-class input τ,

$$[\![K_h]\!](\tau) = [\![K]\!](\mathrm{tr}_h\,\tau).$$

For a drop it is the definition. For an ordinary operation Ψ before the drop, use $\mathrm{tr}_h((\mathscr{E}_{I_h} \otimes \Psi)(\tau)) = \Psi(\mathrm{tr}_h\,\tau)$, by taking the partial trace in a basis of h. This holds for each subnormalized instrument outcome, and tensoring with a reference leaves it valid. Nested recording is an isometry on the ordinary data and the new record, with the older name carried by the identity. Partial traces over distinct names commute at their eliminations. Induction over sequencing, measurement branches and nested bindings proves the formula. Consequently a recording atom R followed by K_h may be replaced by its marginal channel $\rho \mapsto \mathrm{tr}_H(R\rho R^\dagger)$ followed by K. When R is the exact exit record of our body β, this marginal is $\mathsf{while}_c\,\beta$ by (4.2). Under agreement it may also be replaced by the embedded coherent iso. A finite clock has its own marginal and does not in general admit the unconditional exact-record replacement. If a caller is a limit of increasing finite executions with subnormalized outputs, the same pathwise identities preserve its lost trace under that limit. This last observation assumes such a semantics; general recursive callers are outside the finite grammar. □

Why one finite clock does not suffice. The output interface matters. If a caller reads the count, even a fixed count independent of the data is an observable output. Conversely, a finite clock may hide a failure of agreement. Use the permutation $E0 \mapsto E0$ and $E1 \mapsto Bq_1 \mapsto \cdots \mapsto Bq_K \mapsto E1$, with K distinct return values $q_1, \dots, q_K$. Its only nonzero exits are $x_0 = |0\rangle\langle 0|$ and $x_K = |1\rangle\langle 1|$. The K-clock assigns label 0 to both exits. Ignoring it preserves $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, whereas a measurement on every return dephases $|+\rangle$ in that basis. Checking just this clock would accept the replacement incorrectly.

E The derivative and its average

Proof of Theorem 6.1. Work on Y_1. The geometric bound in Theorem B.1 gives an open disk of radius greater than one on which F_z and $\Theta(z)$ are analytic. Empty live space gives $F_z = 0$. The identity $F_z = C + zD_1F_z$ says

$$U_1(u, zF_zu) = (\Theta(z)u, F_zu).$$

Use the wire equation twice, first at (v, w) and then at (u, z). Because $U_1 : X \oplus Y_1 \to X' \oplus Y_1$ preserves inner products, the inner product of the inputs equals that of the outputs. With the inner product linear in its second argument, the two sides are

$$\underbrace{\langle v, u\rangle + \bar{w}z\langle F_wv, F_zu\rangle}_{\text{external and returning inputs}} = \underbrace{\langle \Theta(w)v, \Theta(z)u\rangle + \langle F_wv, F_zu\rangle}_{\text{external and returning outputs}}.$$

Collecting terms gives $I - \Theta(w)^\dagger\Theta(z) = (1-\bar{w}z)F_w^\dagger F_z$. Keep w fixed on the unit circle and differentiate with respect to z. At $z = w$ the factor $1 - \bar{w}w$ is zero, so the term containing F_z' vanishes. The remaining equality is $\Theta(w)^\dagger\Theta'(w) = \bar{w}F_w^\dagger F_w$. Now put $w = e^{i\varphi}$. The angular derivative multiplies $\Theta'(w)$ by iw, giving $Q(w) = F_w^\dagger F_w$ as claimed.

By the telescope in Appendix B, $\sum_{k=1}^{p} kx_k^\dagger x_k = \sum_{j=0}^{p-1} T_j - pT_p$. Geometric decay makes $pT_p \to 0$. Thus $N = \sum_j T_j$. In $F_z^\dagger F_z$ the terms at m, n have factor z^{n-m} on the unit circle. As in Figure 10, integration keeps only the cells where $m = n$. Uniform absolute convergence permits this termwise integration, leaving $\sum_j T_j$. This gives the phase average. For the trace of N, cyclicity changes the summand $\mathrm{tr}\,T_j$ into $\mathrm{tr}(D_1^j CC^\dagger D_1^{\dagger j})$. From the Y_1 output block of $U_1U_1^\dagger = I_{X'\oplus Y_1}$ we have $CC^\dagger =$

$I - D_1 D_1^\dagger$. The partial sum once again telescopes,

$$\sum_{j=0}^{p-1} D_1^j C C^\dagger D_1^{\dagger j} = I_{Y_1} - D_1^p D_1^{\dagger p} \longrightarrow I_{Y_1}.$$

Thus $\operatorname{tr} N = \dim Y_1$. On each unit input u the measured probability of exit k is $\|x_k u\|^2$, so its mean count is $\sum_k k\|x_k u\|^2 = \langle u, Nu\rangle$. □

For the monitored first-return convention the generating function is $z\Theta(z)$, whose relative derivative is $I + Q(z)$. Its mean number of calls is therefore $1 + \langle u, Nu\rangle$ [6, § 3, Eq. (8)]. A phase on the return tag is an ordinary iso, so forming this family requires no counter or guard measurement.

Phase invariance and agreement. On a nonempty external type, the phase family also characterizes agreement. If every exit is $x_k = b_k L$, then $\Theta(z) = f(z)L$ with $|f(z)| = 1$ on the circle, so every marking phase gives the same data channel. Conversely, if those channels are all equal, they equal $\mathscr{E}_L$ because $\Theta(1) = L$. Their average is $\mathscr{M}$ by (6.1), so $\mathscr{M} = \mathscr{E}_L$. Thus agreement says exactly when changing the return phase leaves the data channel unchanged.

F Clause programs and their observations

Many small loop bodies can be written as two clauses with a variable for the payload. We give their definitions here so that one can reproduce each diagram directly from an iso. Each program closes its returning summand with `trace`. We abbreviate `exit` and `back` by E and B in this section's source equations. In these equations B is an injection tag; a matrix block appears only in a denotational equation. In executable types, `x` spells tensor product, and a coefficient `(a :+ b)` denotes $a + ib$ for real a, b.

Two scalar loops in sequence. Use two copies of the scalar guard rotation, with $t = 3/5$ and $s = 4/5$. Their separate returns each receive the phase z, and their external ports connect in sequence. The responses therefore multiply to $f(z)^2$, where $f(z) = (t - z)/(1 - tz)$. Figure 18 gives both this cascade and its single-round matrix W on $\mathbb{C} \oplus (\mathbb{C} \oplus \mathbb{C})$. Multiplying its blocks, or solving its two return equations, gives the same $f(z)^2$. The columns of W give the following three clauses; its two returning basis values are `back 0` and `back 1`.

```
round twoSections : Unit + Qubit = {
  exit unit ↔ [9/25 . exit unit + 4/5 . back 0 + 12/25 . back 1];
  back 0 ↔ [-12/25 . exit unit + 3/5 . back 0 + -16/25 . back 1];
  back 1 ↔ [-4/5 . exit unit + 3/5 . back 1]
}

trace twoSections on unit
```

The command returns $|*\rangle$, since $f(1)^2 = 1$. A period-two clock also returns label 0: both $f(1)^2$ and $f(-1)^2$ are one, so the odd residue sum is zero.

Now carry data u unchanged through both sections and apply a gate L once on the final external output. As Figure 19 shows, this amounts to tensoring the internal evolution with the identity on the data. The response is $f(z)^2 L$. Every exit coefficient multiplies the same L, so dropping the count preserves the data channel by Theorem 5.1. On the unit circle $|f| = 1$, hence the relative derivative of the squared scalar response is twice that of one section. Its phase average gives mean count two, as expected from the mean of one for each rotation. Changing coordinates on the internal space or adjoining an inaccessible summand leaves the exits unchanged.

Two copies of the scalar rotation, $t = 3/5$, $s = 4/5$



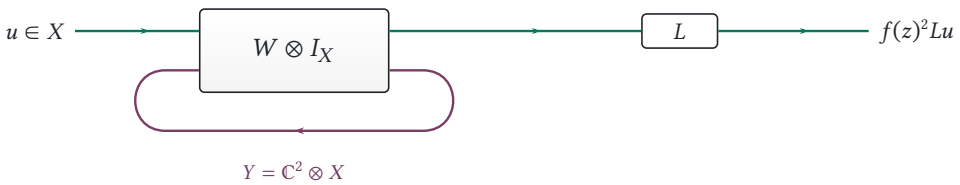
$$f(z) = \frac{t - z}{1 - tz}, \qquad f(1)^2 = 1$$

$$W = \begin{pmatrix} 9/25 & -12/25 & -4/5 \\ 4/5 & 3/5 & 0 \\ 12/25 & -16/25 & 3/5 \end{pmatrix}$$

The same cascade as one body on $\mathbb{C} \oplus (Y_1 \oplus Y_2)$.
Its external-first blocks give $\Theta_W(z) = f(z)^2$.

Fig. 18. Two specified scalar sections give response $f(z)^2$. The numerical matrix is the open body of this cascade in external-first order.

Carry data through this two-section example



$$U = (L \oplus I_Y)(W \otimes I_X), \qquad \Theta_U(z) = f(z)^2 L$$

Here W is the displayed rational two-section body and $L : X \to X$ is unitary. Each exit is a scalar times L. Dropping its count preserves the data channel. The mean number of returns is two on every unit input.

Fig. 19. Tensor the specified two-section body with the data identity and apply L on its external output. This constructed body passes agreement and has mean count two.

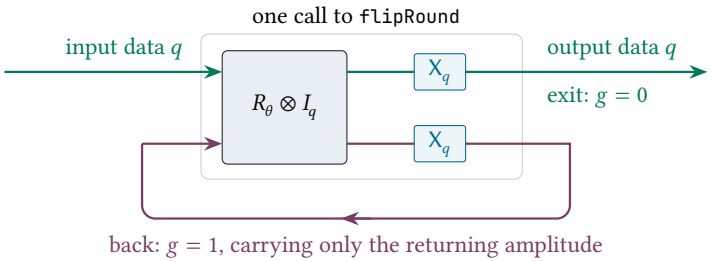
F.1 A flip on every call

For a loop that also transforms a payload on every call, take a guard and a data qubit, apply R_θ to the guard and the data flip $J = \mathsf{X}$ to the data, and close the returning summand. This is more structure than the scalar loop of the main text needs, but it gives a second example where measured and coherent outputs differ. The following source binds the data in each clause before applying its gate.

```
X :: Qubit ↔ Qubit
X = { 0 ↔ 1; 1 ↔ 0 }

round flipRound : Qubit + Qubit = {
  exit x ↔ let y = X x in [ 3/5 . exit y + 4/5 . back y];
  back x ↔ let y = X x in [-4/5 . exit y + 3/5 . back y]
}
```

The first definition names the data gate X inside the box in Figure 20. On input `exit 0` the first clause binds $x = 0$, so `let y = X x` binds $y = 1$. Each dot scales a tagged output, and the brackets form their superposition. The coefficients $3/5$ and $4/5$ are the first column of R_θ, since `exit` means that the guard enters at 0. Following `back 1` into the second clause flips the data once more and uses the other column, with coefficients $-4/5$ and $3/5$. This is where the second call acquired its minus sign. With the guard first and the data second, Figure 21 is the matrix of $R_\theta \otimes \mathsf{X}$.



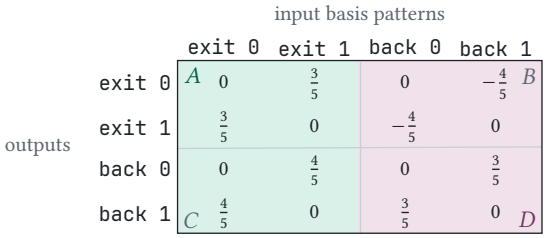
Call	Component entering this call	Leaves the loop	Continues to the next call
1	$\lvert 00\rangle$	$\frac{3}{5}\lvert 01\rangle$	$\frac{4}{5}\lvert 11\rangle$
2	$\frac{4}{5}\lvert 11\rangle$	$-\frac{16}{25}\lvert 00\rangle$	$\frac{12}{25}\lvert 10\rangle$
3	$\frac{12}{25}\lvert 10\rangle$	$-\frac{48}{125}\lvert 01\rangle$	$\frac{36}{125}\lvert 11\rangle$

Fig. 20. The coherent loop connects the body's returning output to its returning input. The two ports are alternative guard values, each carrying the same data qubit q. Both alternatives apply X_q; this data gate commutes with the guard rotation. The table starts at $\lvert 00\rangle$ and writes joint states $\lvert gq\rangle$, with guard first. Emitted components do not enter a later row; only the last column does. No guard measurement is made here.

Which exits does the flip add? Put $t = 3/5$ and $s = 4/5$. The two clauses of `flipRound` give $A = t\mathsf{X}$, $C = s\mathsf{X}$ on entry and $B = -s\mathsf{X}$, $D = t\mathsf{X}$ on return. Hence $x_0 = t\mathsf{X}$ and $x_1 = BC = -s^2 I$. The data flips twice before that second exit. In general $x_{k+1} = -s^2 t^k \mathsf{X}^{k+2}$. Grouping equal data gates gives two

```
exit 0 <-> [3/5 . exit 1 + 4/5 . back 1]
exit 1 <-> [3/5 . exit 0 + 4/5 . back 0]
back 0 <-> [-4/5 . exit 1 + 3/5 . back 1]
back 1 <-> [-4/5 . exit 0 + 3/5 . back 0]
```

one column for each ground input



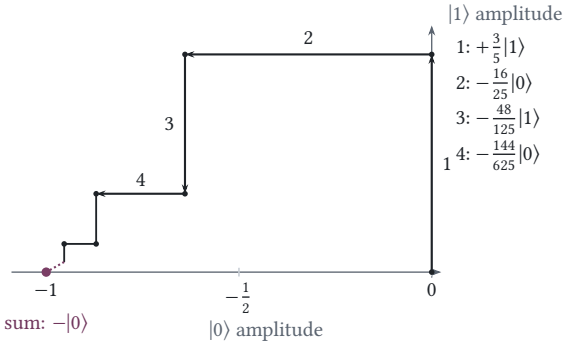
Compare inputs `exit 0` and `back 0`

output row	column product
`exit 1`	$(3/5)(-4/5)$
`back 1`	$(4/5)(3/5)$
inner product	0

The other rows contribute zero.
Each squared norm is $9/25 + 16/25 = 1$.

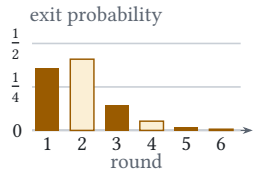
Fig. 21. The four columns of the payload-flipping body, in external-first order. This is $R_\theta \otimes \mathrm{X}$.

(a) `trace`*: add the exit vectors*



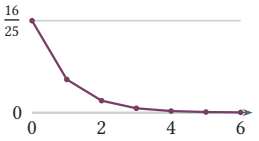
each arrow adds one exit
intermediate points are unnormalized sums

(b) `while`*: add exit probabilities*



odd total: 9/17 on |1⟩
even total: 8/17 on |0⟩

(c) probability still inside



after $k+1$ rounds
each further round retains 9/25

Fig. 22. On input $|0\rangle$, the left panel adds exit amplitudes in the data space. The right panel keeps the exit rounds distinct and plots their probabilities. Calls are numbered from one.

ordinary geometric series,

$$\underbrace{t - s^2 t \sum_{j\geq 0} t^{2j}}_{\text{coefficient of X}} = 0, \qquad \underbrace{-\, s^2 \sum_{j\geq 0} t^{2j}}_{\text{coefficient of } I} = -1, \qquad s^2 = 1 - t^2.$$

Thus the whole coherent loop returns $-u$. In particular it can never return data 1 on input 0. With an exact count the exits that return 0 on this input have odd counts, with amplitudes $-s^2, -s^2t^2, \ldots$. Their squared magnitudes sum to $s^4/(1-t^4) = 8/17$. The complementary probability $9/17$ goes to 1. These are the two observations in Figure 22.

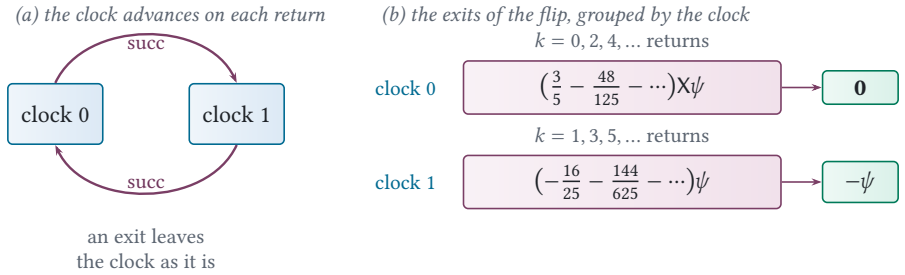


Fig. 23. For the flip, folding by parity cancels all even counts as amplitudes. Measurement of an exact count has positive probability at every count.

The numerical elimination also gives a finite calculation of this loop. Unlike `guardRound`, it has two returning basis patterns. Figure 4 in the main text gives both elimination steps and every new coefficient.

The first substitution in Figure 4 removes `back 0` and leaves coefficient $(3/5)^2 = 9/25$ on the remaining return amplitude y_1. It also changes the coefficient of y_1 in external output v_1 to $(3/5)(-4/5) = -12/25$. On external basis input 0, an amplitude $4/5$ enters this remaining return pattern. Solving its scalar wire equation gives the two entries in that output column:

$$\text{on data } 0: \quad \frac{(4/5)(-4/5)}{1 - 9/25} = -1,$$

$$\text{on data } 1: \quad \frac{3}{5} + \frac{(4/5)(-12/25)}{1 - 9/25} = 0.$$

Interchanging the data bits gives the other column. Thus the finite rewrite produces the same $-I$ that we obtained by adding the exits round by round.

Moving the flip to the exit. A variant that passes the agreement test has the circuit in Figure 24. In the ground clauses, the bit after `exit` is the flipped input bit and the bit after `back` is the input itself.

```
round flipOnExit : Qubit + Qubit = {
  exit 0 ↔ [ 3/5 . exit 1 + 4/5 . back 0];
  exit 1 ↔ [ 3/5 . exit 0 + 4/5 . back 1];
  back 0 ↔ [-4/5 . exit 1 + 3/5 . back 0];
  back 1 ↔ [-4/5 . exit 0 + 3/5 . back 1]
}
```

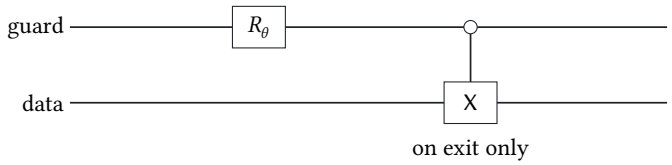


Fig. 24. The same rotation as in `flipRound`, with the flip controlled on the outgoing guard being 0. A returning component keeps its data. Whichever round exits, that component of the data is flipped exactly once.

Running `trace flipOnExit on 0` gives $-|1\rangle$. The measured command `while flipOnExit on 0` gives $|1\rangle\langle 1|$. Its exits are $(3/5)\mathsf{X}$ at count zero and $-(16/25)(3/5)^{k-1}\mathsf{X}$ at counts $k \geq 1$, using the scalar coefficients calculated in Section 3. Every exit is proportional to X, so agreement holds on every input, including one entangled with a reference. The coherent completion is $-\mathsf{X}$.

Rotation with a data iso. For $c = \cos\theta$, $s = \sin\theta$ and an iso $M : a \leftrightarrow a$, define $\mathrm{rotbody}(\theta, M) : a + a \leftrightarrow a + a$ by the clauses

$$E\,x \leftrightarrow \mathsf{let}\ y = M\,x\ \mathsf{in}\ c\,E\,y + s\,B\,y,$$
$$B\,x \leftrightarrow \mathsf{let}\ y = M\,x\ \mathsf{in}\ -s\,E\,y + c\,B\,y.$$

The notation is a schema for named first-order isos, not a higher-order value. Putting $M = \mathsf{X}$ gives the four ground instances of the flip. For $|c| < 1$ its exits are cM and $-s^2c^kM^{k+2}$. Summing them gives

$$f_\theta(M) = M(cI - M)(I - cM)^{-1}.$$

On an eigenvector of M with eigenvalue w, the equation $f_\theta(w) = 1$ reduces to $w^2 - 2cw + 1 = 0$, while $f_\theta(w) = -1$ reduces to $w^2 = 1$. Hence taking M to be the real rotation of angle $-\theta$ gives the identity. When $|c| = 1$, the guard never enters and the immediate exit gives the same conclusion for that rotation. Taking $M = \mathsf{X}$ gives $-I$ for $|c| < 1$.

Here is the inverse-rotation instance at $c = 3/5$, including its invocation. The first declaration supplies $M = R_{-\theta}$; the second applies it locally on each call.

```
inverseRotation :: Qubit ↔ Qubit
inverseRotation = {
  0 ↔ [3/5 . 0 + -4/5 . 1];
  1 ↔ [4/5 . 0 + 3/5 . 1]
}

round undoRound : Qubit + Qubit = {
  exit x ↔ let y = inverseRotation x in
    [3/5 . exit y + 4/5 . back y];
  back x ↔ let y = inverseRotation x in
    [-4/5 . exit y + 3/5 . back y]
}

trace undoRound on 0
```

The command returns $|0\rangle$. On an arbitrary vector the completed gate is I, as the eigenvalue calculation shows. The measured loop need not agree: on $|+\rangle$ it gives $\frac{1}{650}\begin{pmatrix} 349 & -107 \\ -107 & 301 \end{pmatrix}$ rather than $|+\rangle\langle +|$. The cancellation giving the coherent identity takes place between exits on different rounds.

A guard whose two payloads differ. A tag test need not be a measurement of a qubit register beside an unchanged data type. Take the two external patterns 0, 1 and just one return pattern $*$. The body below is the three-cycle of monitored recurrence, with its basis relabeled [6, Example 3.1]. It binds no history variable.

```
cycle3 :: (Qubit + Unit) ↔ (Qubit + Unit)
cycle3 = {
  left (left unit)  ↔ right unit;
  left (right unit) ↔ left (left unit);
  right unit        ↔ left (right unit)
}
```

Input 1 exits immediately as 0, whereas input 0 returns once and exits as 1. Thus $x_0 = |0\rangle\langle 1|$, $x_1 = |1\rangle\langle 0|$, and all later exits vanish. Calling `trace` on this iso gives the flip X, so $|+\rangle$ returns unchanged. Calling `count` with period two gives $(|0\rangle|0\rangle + |1\rangle|1\rangle)/\sqrt{2}$, with counter first. Calling `while` on the very same iso returns $I/2$ on that input. The two exit vectors on $|+\rangle$ have zero inner product, but they are values of the data we retain. Coherent feedback still preserves the terms $|0\rangle\langle 1|$ and $|1\rangle\langle 0|$ between them. A separate ignored count removes these terms because it gives the two data vectors different orthogonal labels. Thus distinguishable data values alone do not make recording the round harmless.

A cycle, a reflection and a phase. The cycle3 program just defined gives the data flip after closing its one return pattern. One may add a dead pattern with its own identity clause, obtaining the same flip on a two-pattern return type. A body with no data can instead be the reflection $E* \leftrightarrow t\,E* + s\,B*$, $B* \leftrightarrow s\,E* - t\,B*$ at $t = 3/5, s = 4/5$. Its immediate exit is t, and its delayed sum is $s^2/(1+t)$. They add to 1.

```
round reflection : Unit + Unit = {
  exit unit ↔ [3/5 . exit unit + 4/5 . back unit];
  back unit ↔ [4/5 . exit unit + -3/5 . back unit]
}

trace reflection on unit
```

This command returns $|*\rangle$. The measured loop also returns the sole external value with probability one. Its parity record is $|1\rangle \otimes |*\rangle$: the even exits cancel and the odd exits add to one. This reflection and `guardRound` therefore have the same measured data output but opposite coherent phases.

For another example, take the four permutation clauses

$$E0 \mapsto E0, \quad E1 \mapsto B1, \quad B0 \mapsto B0, \quad B1 \mapsto E1.$$

Multiply every output by g with $|g| = 1$. Input 0 receives one factor of g and input 1 receives two, giving $\mathrm{diag}(g, g^2)$. The one-call channel is unchanged by this common phase but the loop channel in general changes. Placing a phase z on the *returning* outputs alone instead gives $\mathrm{diag}(1, z)$. The markings occur in different places, but at $z = g$ their completions differ only by the global factor g. They therefore give the same data channel for this permutation. That channel can still differ from the unmarked identity channel.

The following two programs instantiate $g = z = i$, written `(0 :+ 1)` in executable syntax. Both use the exact input $3|0\rangle/5 + 4|1\rangle/5$.

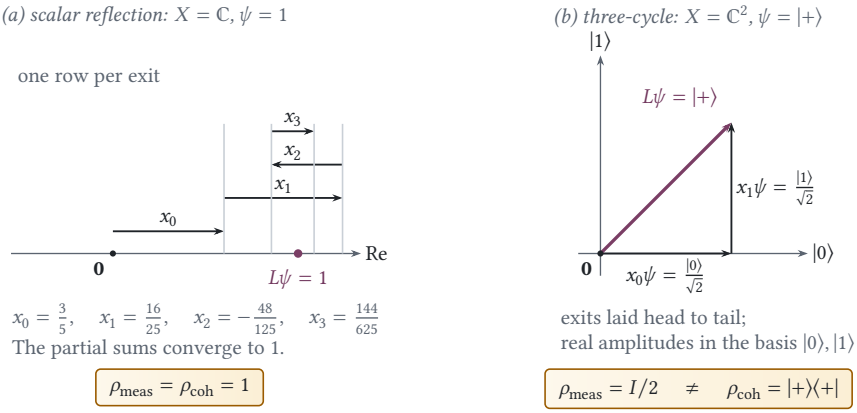


Fig. 25. The scalar reflection gives the same data under measurement or coherent feedback. For the three-cycle, measuring yields $I/2$ while coherent feedback returns the pure state $|+\rangle$.

```
round unmarked : Qubit + Qubit = {
  exit 0 ↔ exit 0;
  exit 1 ↔ back 1;
  back 0 ↔ back 0;
  back 1 ↔ exit 1
}

round wholePhase : Qubit + Qubit = {
  exit 0 ↔ (0 :+ 1) . exit 0;
  exit 1 ↔ (0 :+ 1) . back 1;
  back 0 ↔ (0 :+ 1) . back 0;
  back 1 ↔ (0 :+ 1) . exit 1
}

trace wholePhase on 3/5 . 0 + 4/5 . 1
```

This invocation returns $3i|0\rangle/5 - 4|1\rangle/5$.

```
round returnPhase : Qubit + Qubit = {
  exit 0 ↔ exit 0;
  exit 1 ↔ (0 :+ 1) . back 1;
  back 0 ↔ (0 :+ 1) . back 0;
  back 1 ↔ exit 1
}

trace returnPhase on 3/5 . 0 + 4/5 . 1
```

The second returns $3|0\rangle/5 + 4i|1\rangle/5$. Multiplying it by i gives the first output, so their density operators agree.

Kickback with a returned ancilla. Let $J = \mathsf{X}$, H be Hadamard, $G = HJ$, and $\Pi_\pm = (I \pm J)/2$. Use rotbody$(\pi/4, M)$ with ancilla and data as payload, where $M = G \otimes \Pi_+ + JG \otimes \Pi_-$. In iso syntax this M is an operation controlled in the J basis of the data: change that basis with H, apply G or

JG to the ancilla according to the resulting tag, then undo the basis change. In the first sector $G^2 = \sqrt{2}G - I$, so $f_{\pi/4}(G) = I$. In the second $(JG)^2 = I$, so $f_{\pi/4}(JG) = -I$. The completed loop is $I \otimes (\Pi_+ - \Pi_-) = I \otimes J$. Thus it returns the ancilla on arbitrary joint inputs, not only on a selected initialization.

In the basis of the two payload bits, the projector sum gives

$$M|0,d\rangle = \frac{|0,1-d\rangle - |1,1-d\rangle}{\sqrt{2}}, \qquad M|1,d\rangle = \frac{|0,d\rangle + |1,d\rangle}{\sqrt{2}}.$$

The data bit flips in the first case and stays in the second. Expanding the guard rotation now gives the clauses below. Each nonzero coefficient of M is $\pm 1/\sqrt{2}$, and the guard contributes another $1/\sqrt{2}$, leaving only coefficients $\pm 1/2$. Thus no rounded Hadamard coefficient is needed in this source.

```
round kickbackExact : (Qubit x Qubit) + (Qubit x Qubit) = {
  exit ⟨0, 0⟩ ↔ [1/2 . exit ⟨0, 1⟩ + -1/2 . exit ⟨1, 1⟩
                 + 1/2 . back ⟨0, 1⟩ + -1/2 . back ⟨1, 1⟩];
  exit ⟨0, 1⟩ ↔ [1/2 . exit ⟨0, 0⟩ + -1/2 . exit ⟨1, 0⟩
                 + 1/2 . back ⟨0, 0⟩ + -1/2 . back ⟨1, 0⟩];
  exit ⟨1, 0⟩ ↔ [1/2 . exit ⟨0, 0⟩ + 1/2 . exit ⟨1, 0⟩
                 + 1/2 . back ⟨0, 0⟩ + 1/2 . back ⟨1, 0⟩];
  exit ⟨1, 1⟩ ↔ [1/2 . exit ⟨0, 1⟩ + 1/2 . exit ⟨1, 1⟩
                 + 1/2 . back ⟨0, 1⟩ + 1/2 . back ⟨1, 1⟩];
  back ⟨0, 0⟩ ↔ [-1/2 . exit ⟨0, 1⟩ + 1/2 . exit ⟨1, 1⟩
                 + 1/2 . back ⟨0, 1⟩ + -1/2 . back ⟨1, 1⟩];
  back ⟨0, 1⟩ ↔ [-1/2 . exit ⟨0, 0⟩ + 1/2 . exit ⟨1, 0⟩
                 + 1/2 . back ⟨0, 0⟩ + -1/2 . back ⟨1, 0⟩];
  back ⟨1, 0⟩ ↔ [-1/2 . exit ⟨0, 0⟩ + -1/2 . exit ⟨1, 0⟩
                 + 1/2 . back ⟨0, 0⟩ + 1/2 . back ⟨1, 0⟩];
  back ⟨1, 1⟩ ↔ [-1/2 . exit ⟨0, 1⟩ + -1/2 . exit ⟨1, 1⟩
                 + 1/2 . back ⟨0, 1⟩ + 1/2 . back ⟨1, 1⟩]
}

trace kickbackExact on 3/5 . ⟨0, 0⟩ + 4/5 . ⟨1, 1⟩
```

The input is entangled between ancilla and data. Its output is $3|0,1\rangle/5 + 4|1,0\rangle/5$, exactly the result of applying J to the second component. The same identity on arbitrary joint inputs follows from the two-sector calculation above.

F.2 Other retained histories

In the map example of *From Symmetric Pattern-Matching to Quantum Control*, the iso argument $g : a \leftrightarrow b$ is applied to a head and the recursive iso f to the tail. In the source's list notation the two clauses read [19, § 4]

$$[] \leftrightarrow [],$$
$$h : t \leftrightarrow \text{ let } x = g\,h \text{ in let } y = f\,t \text{ in } x : y.$$

If g is Hadamard and the input has length two, every output of the recursion still has length two. A superposition of input lengths two and three likewise produces two components that can be distinguished by length alone. This explains how a reversible program can retain the depth of its recursion without ever measuring it. The cyclic register in (4.1) lets us make the same distinction, up to its period, on a body whose input need not contain a list.

An exact record may also be carried by orthogonal time slots: one tick of delay on the return wire puts $x_0u, x_1u, \ldots$ in consecutive output slots [3, § 3.4]. In the common-clock halting construction, a common post-halt ancilla sequence likewise distinguishes halt times at a fixed observation time. The original model also allows different ancilla sequences across branches, which can restore interference later [16]. Our exact record requires distinct halt times to remain orthogonal at observation. Appendix H gives the corresponding short-pulse and single-frequency experiments.

A case inside or outside feedback. Let θ have $\cos\theta = t = 3/5$ and rotate the guard alone. A phase z on the returning output gives $\theta_0(z) = (t - z)/(1 - tz)$ by summation of the scalar exits. The case $I \oplus \mathrm{trace}(\mathrm{rotbody}(\theta, I); (I \oplus zI))$ on a reference qubit therefore denotes $\mathrm{diag}(1, \theta_0(z))$. If instead the data bit selects the phase on a return *inside* the guard rotation, the two data sectors get $\theta_0(1)$ and $\theta_0(z)$. The result is $\mathrm{diag}(-1, \theta_0(z))$. The bypass in the first program makes no visit to the loop, whereas both sectors in the second visit it. At $z = i$, the two open bodies are the following literal programs. A factor `(0 :+ 1)` contributes i only to a returning output.

```
round phaseOutside : Qubit + Unit = {
  exit 0 ↔ exit 0;
  exit 1 ↔ [3/5 . exit 1 + (0 :+ 1) . 4/5 . back unit];
  back unit ↔ [-4/5 . exit 1 + (0 :+ 1) . 3/5 . back unit]
}

trace phaseOutside on 3/5 . 0 + 4/5 . 1

round phaseInside : Qubit + Qubit = {
  exit 0 ↔ [3/5 . exit 0 + 4/5 . back 0];
  exit 1 ↔ [3/5 . exit 1 + (0 :+ 1) . 4/5 . back 1];
  back 0 ↔ [-4/5 . exit 0 + 3/5 . back 0];
  back 1 ↔ [-4/5 . exit 1 + (0 :+ 1) . 3/5 . back 1]
}

trace phaseInside on 3/5 . 0 + 4/5 . 1
```

Here $\theta_0(i) = (15 - 8i)/17$. On the stated input the first returns $3|0\rangle/5 + (12/17 - 32i/85)|1\rangle$, while the second negates only the $|0\rangle$ coefficient. Thus the position of the case changes the relative phase between the two data alternatives. On $|+\rangle$ followed by H, their probabilities of outcome 1 are $1/17$ and $16/17$, respectively.

G Reading the feedback diagrams

The sheet diagrams read from bottom to top. Each summand has a separate surface. Looking along the depth of one surface reveals the wires of its tensor factors, so two registers on one sheet can be used together, while different sheets carry alternative components. The dark segment at which surfaces meet is a seam of the body. Closing a summand bends its entire surface back to the input. This last bend denotes the trace proved here. The underlying sheet calculus describes sum, tensor and composition [8, Defs. 16, 18–20]. A trace in the sense of this paper closes an entire summand. For instance the ordinary partial matrix trace of controlled-X over the target gives $2|0\rangle\langle 0|$. For the summand trace in Figure 26, reorder the tensor factors to group by the target c, which carries the exit or return tag. The control x remains the external data. Control 0 leaves c unchanged and exits immediately. Control 1 flips c twice and exits after one return. The exit operators are $|0\rangle\langle 0|$ and

$|1\rangle\langle 1|$, whose sum is the identity on the control. These computations even have different types of interpretation.

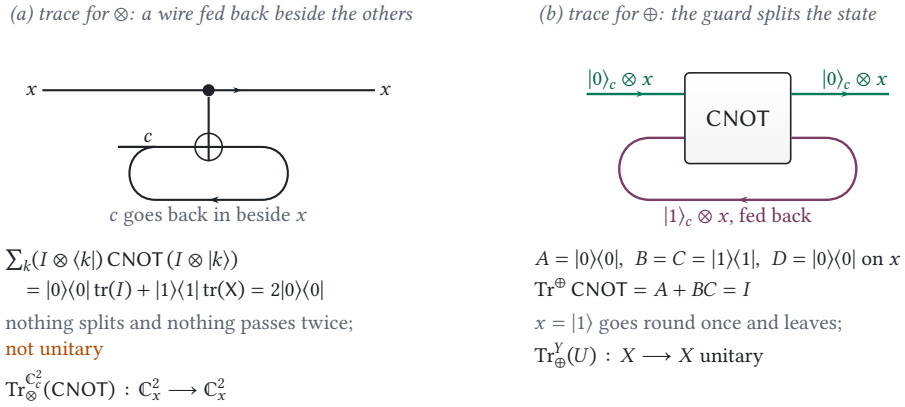


Fig. 26. A tensor-factor partial trace and a summand feedback trace of controlled-X evaluate different diagrams.

G.1 The part of the return space that an input can reach

Figures 27 and 28 show the invariant-space argument and its finite decay bound. There is also a finite way to identify Y_0 without diagonalizing D. Set $K_0 = Y$ and $K_{j+1} = \ker B \cap D^{-1}(K_j)$. Then K_j is the set of vectors with their first j emissions zero. An equality $K_j = K_{j+1}$ makes this space invariant and fixes all subsequent terms; strictly decreasing steps can occur at most $\dim Y$ times. Consequently $K_m = Y_0$ for $m = \dim Y$. When $m > 0$ and $Y_1 \neq 0$, the norm of D_1^m is strictly less than one: a unit vector achieving norm one would have emitted nothing in the first m rounds and would lie in Y_0. Compactness supplies a maximum, hence a uniform geometric bound. This gives another proof of the decay used in Appendix B.

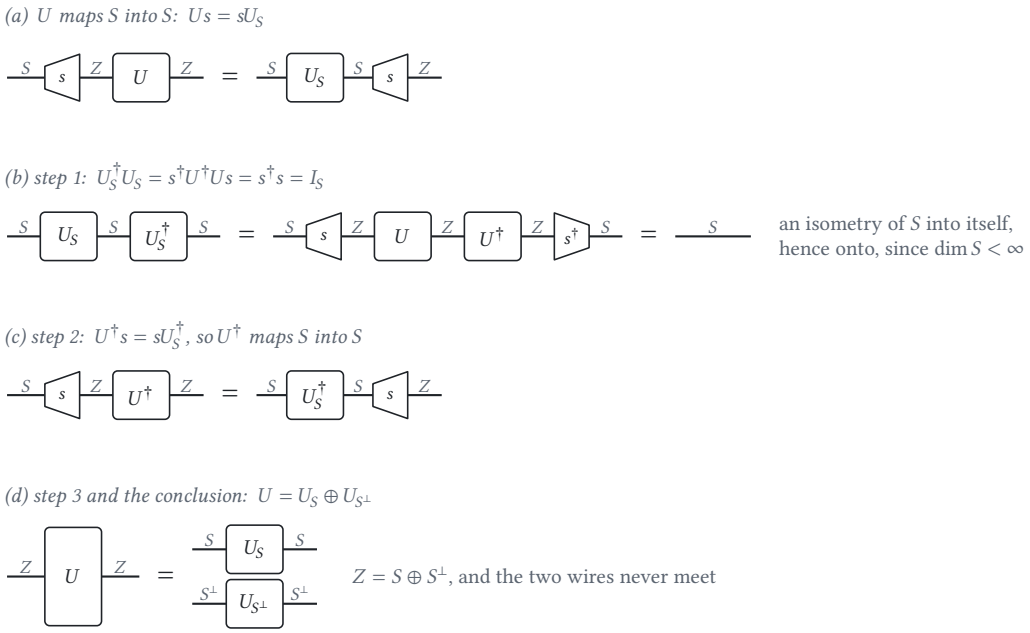


Fig. 27. For a unitary $U : Z \to Z$, let $s : S \rightarrowtail Z$ be the inclusion of a finite invariant subspace. Norm preservation gives an isometry on S, which is onto in finite dimension; hence S reduces U.

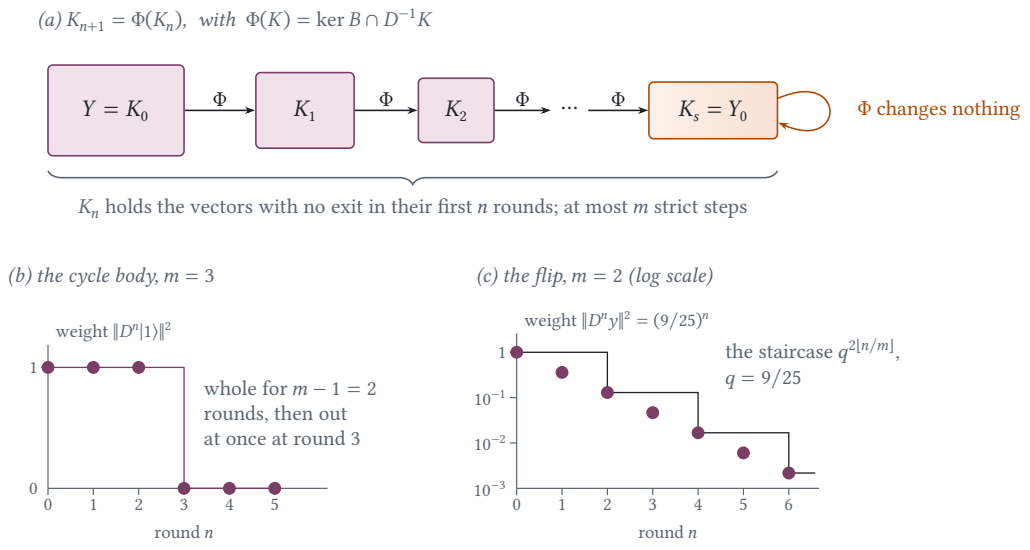


Fig. 28. The no-emission spaces stabilize within $\dim Y$ steps. Panel (b) is the four-state cycle $0 \mapsto 1 \mapsto 2 \mapsto 3 \mapsto 0$, with external state 0. Its three-dimensional return block has $D|1\rangle = |2\rangle$, $D|2\rangle = |3\rangle$ and $D|3\rangle = 0$. In panel (c), $\|D_1\| = 3/5$ and $m = 2$, so $q = \|D_1^2\| = 9/25$. The exact residual weight is $(9/25)^n$, and the staircase bounds it by $q^{2\lfloor n/m \rfloor}$.

G.2 Effects and trace laws

Figure 12 places the three readings in the Π family. Figure 29 adds the partial-injection model and the compact completion. The finite counter is obtained by extending the body before tracing. Figure 30 shows why coherent sum does not descend to channels. The trace on finite isometries agrees on embedded unitaries with the original coherent trace [3, Cor. 3.3.16], so allocation alone does not turn coherent feedback into measurement. The exact countable record is a semantic limit of the finite recording programs. Measuring the tag instead uses the classical sum of operator blocks. All three constructions start from the same iso body.

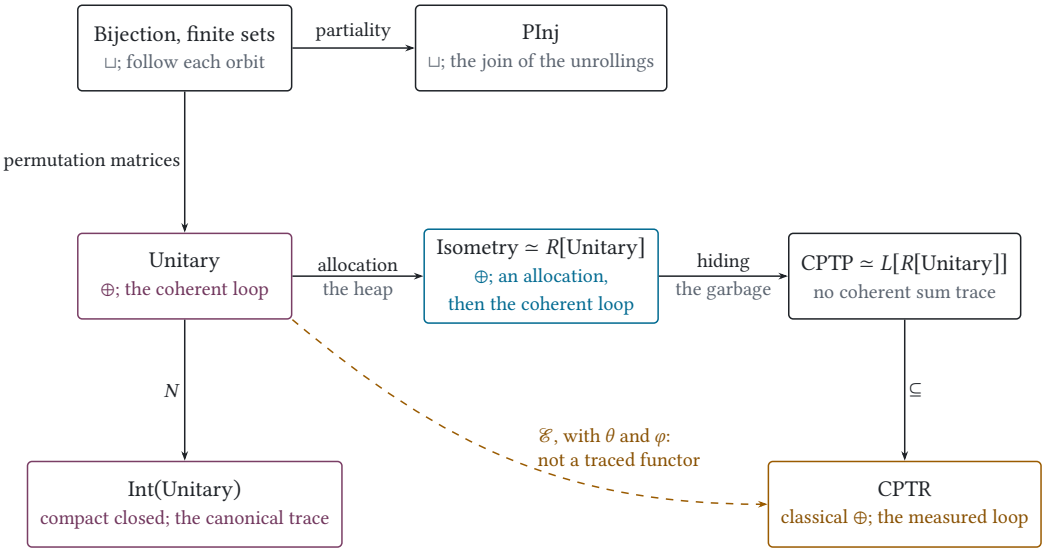


Fig. 29. The measured loop uses a classical sum of operator algebras. The dashed body translation is $U \mapsto \Phi_U = \varphi \circ \mathscr{E}_U \circ \theta$. The direct-sum trace of an embedded unitary remains coherent on the isometry layer.

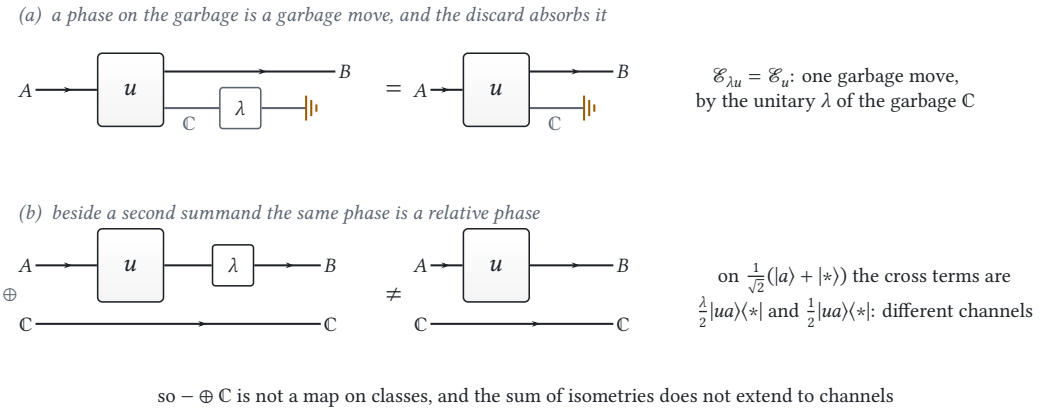


Fig. 30. Here $u : A \rightarrowtail B$ is an isometry, $A \neq 0$, and $|\lambda| = 1$ with $\lambda \neq 1$. A one-dimensional garbage factor absorbs this phase. Putting u in one arm of a coherent sum exposes it instead.

G.3 Composition and the backward wire

The summand matters. In a traced monoidal category the trace closes one component of a morphism $X \odot Y \to X' \odot Y$, for the category's chosen product $\odot$ [13]. Here it is $\oplus$. The usual tensor trace on finite linear maps is a different operation and need not send a unitary to a unitary: tracing the factor $\mathbb{C}^2$ out of $I_X \otimes I_{\mathbb{C}^2}$ gives $2I_X$.

Consider the iso swapping external and internal copies of a (Figure 34). Its input makes one return and comes out unchanged, so its coherent trace is the identity: the law called *yanking*. A K-clock with $K > 1$ distinguishes this path from an immediate exit by labels 1 and 0. Naturality and sliding move external and internal operations across a trace, as drawn in Figure 33 [4, 13]. Figure 12 follows three further equations, with canonical reassociations understood. Closing two return sheets successively is the same as closing their sum. A bypass stays outside the closure. In sequence, the first loop's external output supplies the second, but each return stays with its own body. Writing their exit families as x_i^U, x_j^V, the combined body emits $\sum_{i+j=k} x_j^V x_i^U$ at total count k. These are equations of coherent feedback at the displayed external types.

Writing the guard as a tensor factor and applying distributivity gives exactly the two injection tags in a clause (Figure 31). Naturalities of the additive trace apply to this representation. Closing the body $I \oplus U$ over U's return port gives $I \oplus L$: the immediate exit is $I \oplus x_0$ and the later exits are $0 \oplus x_k$. Closing disjoint internal ports successively gives the simultaneous solution by the argument of Appendix B.1, as drawn in Figure 32. For two loops in sequence, at a common phase z their responses multiply as $\Theta_2(z)\Theta_1(z)$, because the external output of the first enters the second and their return ports are separate. Expanding the product gives the convolution $\sum_{j=0}^{k} x_{k-j}^{(2)} x_j^{(1)}$ at total count k. The adjoint of a coherent loop is the loop of the adjoint body: its blocks give exactly $x_k^\dagger$ as exits and the series is absolutely convergent (Figure 35).

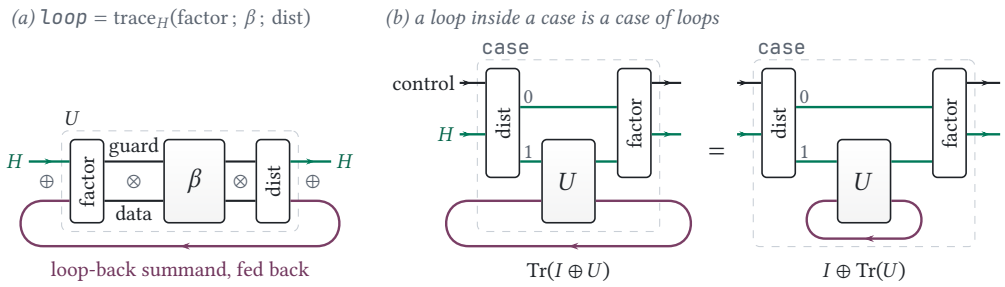


Fig. 31. Distributivity makes the guard a summand. A case with a separate untouched arm distributes over the trace of the other arm.

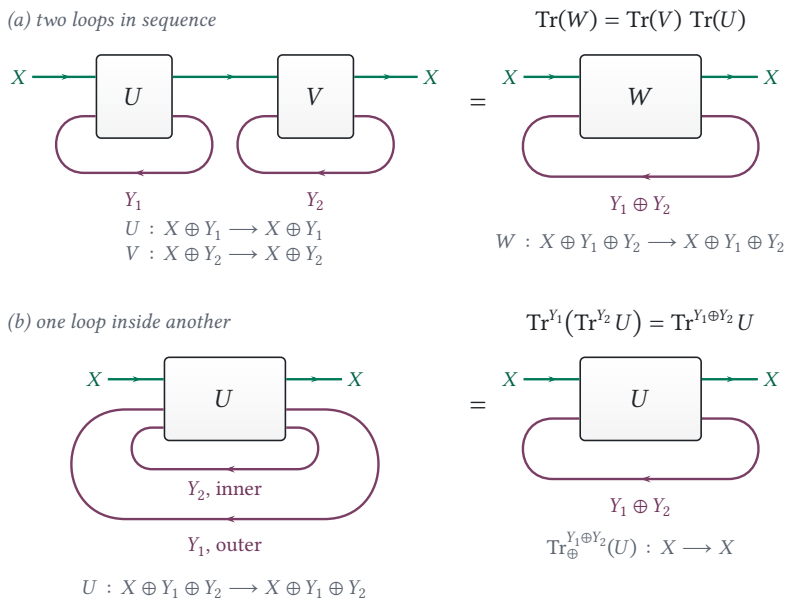
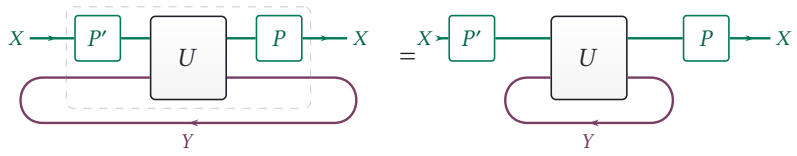


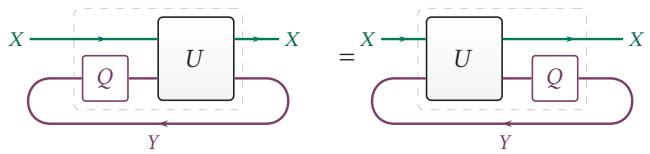
Fig. 32. Sequencing convolves the exit operators; nested closure over internal ports solves their feedback equations jointly.

(a) naturality



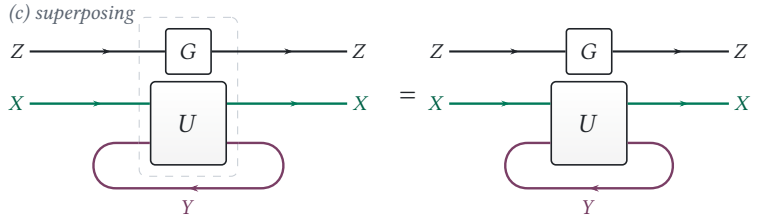
$$\mathrm{Tr}^{Y}\big((P \oplus I)\,U\,(P' \oplus I)\big) = P\ \mathrm{Tr}^{Y}(U)\,P'$$

(b) sliding



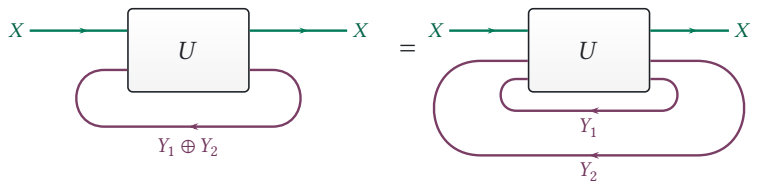
$$\mathrm{Tr}^{Y}\big(U\,(I \oplus Q)\big) = \mathrm{Tr}^{Y}\big((I \oplus Q)\,U\big)$$

(c) superposing



$\mathrm{Tr}^{Y}(G \oplus U) = G \oplus \mathrm{Tr}^{Y}(U)$, the loop taken over Y

(d) vanishing



$$\mathrm{Tr}^{Y_1 \oplus Y_2}(U) = \mathrm{Tr}^{Y_2}\big(\mathrm{Tr}^{Y_1}(U)\big)$$

Fig. 33. Naturality, sliding, superposing and vanishing are typed equations for coherent loops. In panel (a), $P, P' : X \to X$.

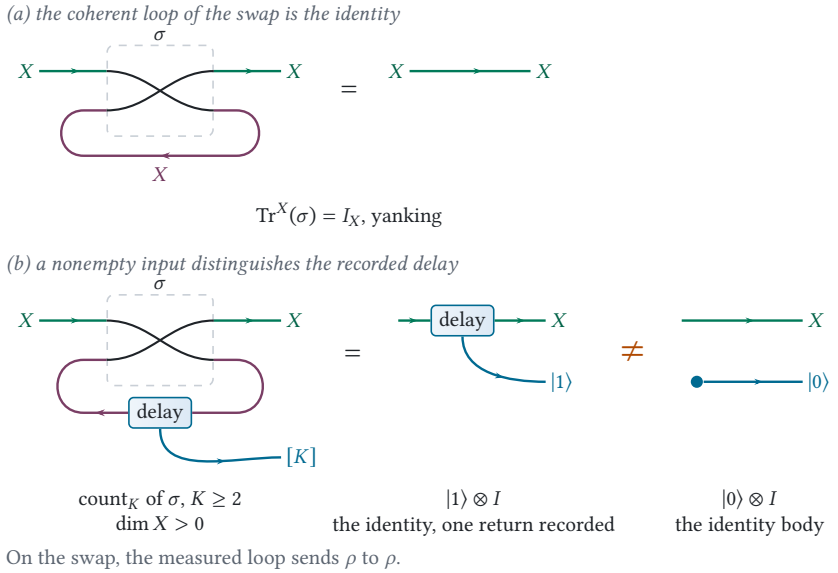


Fig. 34. The swap is between two copies of X. Its coherent trace returns the input. For $\dim X > 0$ and $K \geq 2$, a recording loop distinguishes one return from an immediate exit.

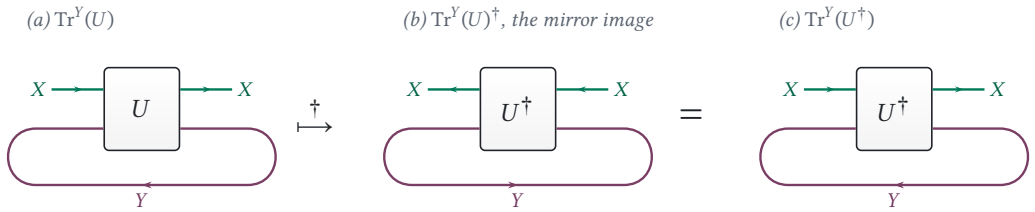


Fig. 35. Reversing a coherent body exchanges entry and exit and takes the adjoint of each exit amplitude.

In the loop diagram a wire runs backwards from the returning output to the next input. To make this drawing into a typed composition, the Int construction pairs each forward type P^+ with a backward type P^-. A map from P to Q can then consume P^+ from its left and Q^- from its right, so its underlying unitary has type $P^+ \oplus Q^- \to Q^+ \oplus P^-$. Figure 37 draws these ports. Composition connects the middle ports by feedback, as in Figure 38. In particular, an ordinary space enters the construction as $N(X) = (X, 0)$, with no backward input. Bending Y around an embedded body identifies the amplitude leaving it, $Cu + Dy$, with the amplitude y supplied on the same wire. The compact closed trace drawn in Appendix G, Figure 36, therefore returns exactly $N(L)$, by the very wire equation used in Theorem B.1 [13] [17, Thms. 5.15–5.16]. Inserting a successor changes what comes back along that wire. Measuring the tag changes what the wire carries from an amplitude to a classical alternative.

The sum in Int reverses the order of negative components:

$$(P^+, P^-) \oplus (Q^+, Q^-) = (P^+ \oplus Q^+, Q^- \oplus P^-).$$

The dual of $(Y, 0)$ is $(0, Y)$, and the unit and counit are identity wires on Y with their ports rearranged. Composing two maps plugs the forward and backward middle ports together and takes the original

trace. With $\eta_{NY} : 0 \to (-Y) \oplus NY$ and $\varepsilon_{NY} : NY \oplus (-Y) \to 0$, the typed closure is

$$N(\mathrm{Tr}^Y U) = (I_{NX'} \oplus \varepsilon_{NY})(NU \oplus I_{-Y})(I_{NX} \oplus \sigma_{-Y,NY})(I_{NX} \oplus \eta_{NY}).$$

The swap puts the cup's two outputs in the order required by $NU \oplus I_{-Y}$. Its underlying map is $I_{Y \oplus Y}$, since both source and target pairs are (Y, Y). With the external identity this is $I_{X \oplus Y \oplus Y}$, explaining the second 1 in Figure 36(b). The four regions between the three dashed slices in Figure 36(b) show these four factors in execution order. If two perfect reflectors enclose a mode, that mode need not have a unique internal amplitude, but it has a unique external response by Theorem B.1. A finite clock bends the enlarged internal type $[K] \otimes Y$ and advances its label on each passage. These statements require no tensor-product duals of unitary gates themselves. Figure 39 separates these constructions. Under direct sum a cup from 0 to $Y \oplus Y^*$ cannot be unitary unless $Y = 0$. Under tensor, a Hilbert-space cup and cap obeying the snake identity on an n-dimensional object have product of norms at least n: closing to a scalar gives n and Cauchy–Schwarz bounds it by the product. Thus for $n \geq 2$ they cannot both be contractions. The compact completion uses the sum, where its objects are pairs.

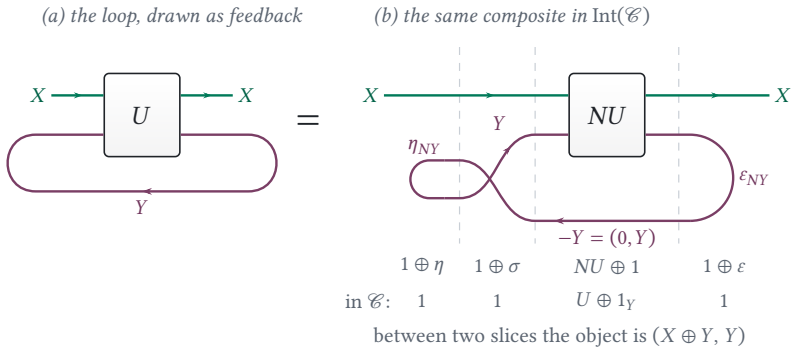


Fig. 36. For $X' = X$ the negative port in the compact closed completion connects the output of the return summand to its input. The same amplitude must satisfy the feedback equation at both ends.

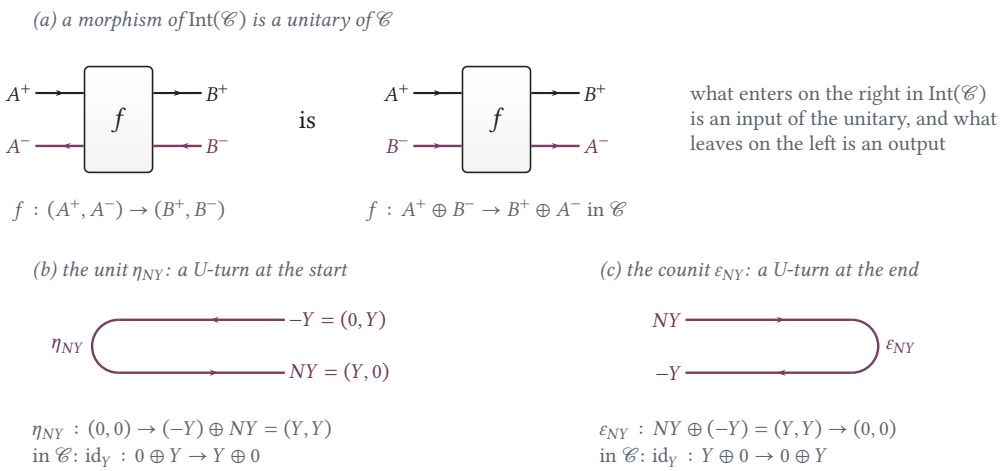
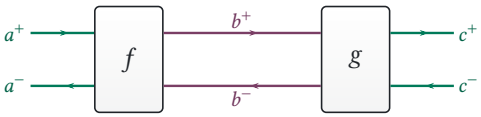


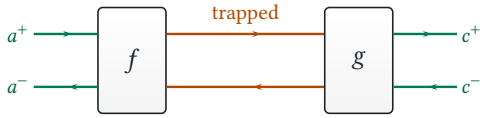
Fig. 37. Objects in the compact completion have forward and backward components. Cups and caps bend a port within that completion.

(a) $g \circ f$: a loop over the middle wires



each round trip from f to g and back
multiplies by $f_{12}\, g_{21}$; the composite adds them

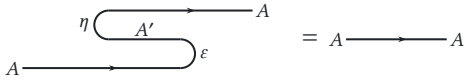
(b) two perfect mirrors: $f_{12}\, g_{21} = I$



the light between them is a dead part;
$g \circ f$ is the outer reflection of each mirror

Fig. 38. With rows and columns in the displayed order, $f_{12} : B^- \to B^+$ and $g_{21} : B^+ \to B^-$. Thus $b^+ = f_{11}a^+ + f_{12}b^-$ and $b^- = g_{21}b^+ + g_{22}c^-$ give round trip $f_{12}g_{21}$ on B^+. A perfectly reflecting interior is a dead summand invisible from outside.

(a) the tensor product: a dual would make the loop of the identity the dimension

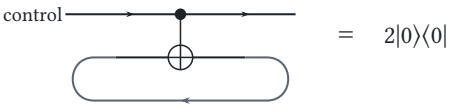


the snake equation defines the dual



so $\varepsilon\, \sigma\, \eta = \dim A$, while $|\varepsilon\, \sigma\, \eta| \le \|\varepsilon\|\, \|\eta\| \le 1$
among contractions: no dual once $n \ge 2$

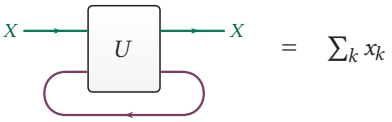
(b) the trace of a factor adds the blocks of one application



tr over the target, a factor

$(I \otimes \langle 0|)\,\mathrm{CNOT}\,(I \otimes |0\rangle) + (I \otimes \langle 1|)\,\mathrm{CNOT}\,(I \otimes |1\rangle)$
$= |0\rangle\langle 0| + |0\rangle\langle 0|$, not a contraction

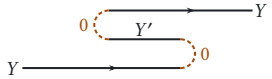
(c) the trace of a summand adds the exits of the rounds



the loop-back summand Y

$y = Cu + Dy$ implies
$U(u, y) = (\mathrm{Tr}^Y(U)u, y)$,
so $\|\,\mathrm{Tr}^Y(U)u\| = \|u\|$

(d) the direct sum: the unit and counit through the zero space are zero maps

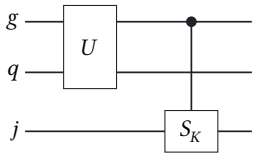


not the identity of Y: every map into or out of the zero space is zero,
and a unitary into or out of it exists only when $Y = 0$

Fig. 39. The compact completion supplies the negative type. The original tensor product on contractions does not supply a dual of dimension two or greater.

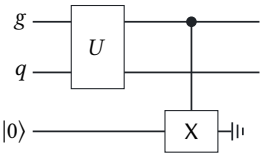
Figure 40 puts the counter increment and the tag-copying step beside the same body.

Keep the count modulo K



$S_K|j\rangle = |j + 1 \bmod K\rangle$.

Measure only the guard tag



Discard the fresh bit after this call.

Fig. 40. One call to the same body U, followed by an operation controlled on the outgoing guard $g = 1$. A filled dot denotes a coherent control. The left circuit carries j into the next round. In the right circuit the discarded bit distinguishes the tags without distinguishing states of the payload q. Bare crossings have no gate.

G.4 The same operations on sheets

The finite counter in Figure 6 uses the depth of the page. The successor acts on its counter wire and leaves the payload wire alone. Figures 8 and 10 place the measurement and phase on that return. Figure 12 shows how separate return sheets compose, in the same upward orientation. In particular a seam labeled U acts on its incident summands even when one drawing of the external wire is partially hidden by the returning surface.

H A ring realization of the same exits

The reference circuit in Section 3 makes the scalar phase observable. The same exits arise from an ideal lossless coupler between an external waveguide mode and a circulating ring mode. Its through amplitude is $t = 3/5$ and its crossing amplitude is, where $s = 4/5$. Figure 41 gives its clauses and feedback equation.

The coupler is $(1 \oplus i)R_\theta(1 \oplus (-i))$. Changing the returning basis in this way gives entrance and exit coefficients both equal to is, with return coefficient t. Their product on exit $k + 1$ is $(is)t^k(is) = -s^2t^k$, so every external exit is unchanged. With phase z per circuit, its single-frequency response is therefore $\Theta_{\text{ring}}(z) = (t - z)/(1 - tz)$, as for the rotation. The same coupler has a literal rational-complex implementation:

```
round ringCoupler : Unit + Unit = {
  exit unit ↔ [3/5 . exit unit + (0 :+ 1) . 4/5 . back unit];
  back unit ↔ [(0 :+ 1) . 4/5 . exit unit + 3/5 . back unit]
}

trace ringCoupler on unit
```

The command returns $-|*\rangle$, as does the guard rotation. The returning basis change preserves every exit, not only their sum.

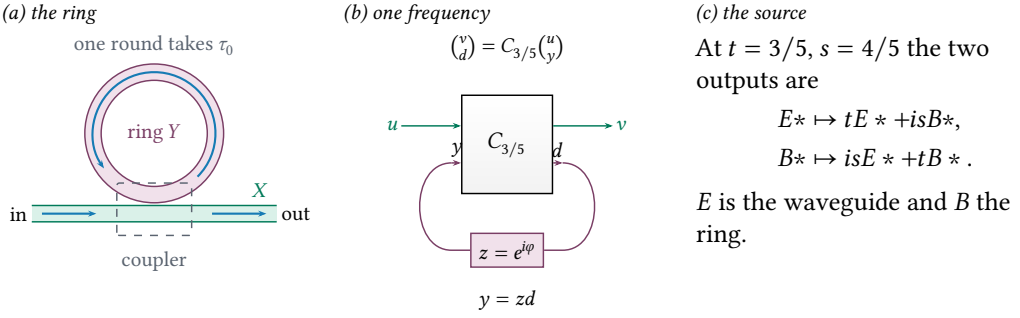


Fig. 41. The ring, the feedback equation at a single frequency, and the two clauses of the coupler. Here d is the amplitude leaving the coupler inside the ring, and $y = zd$ enters it again.

In this lossless coupler [2, Eqs. (6)–(7)], the observation depends on the input pulse. If the pulse lasts less than one circuit time, the outgoing pulse on each round ends before the next begins. Measuring the arrival time then distinguishes the exits and gives their probabilities. At a single frequency all the returning waves occupy the same mode and their amplitudes add [18, Eqs. (32)–(33)]. Figure 42 draws these two mode specifications.

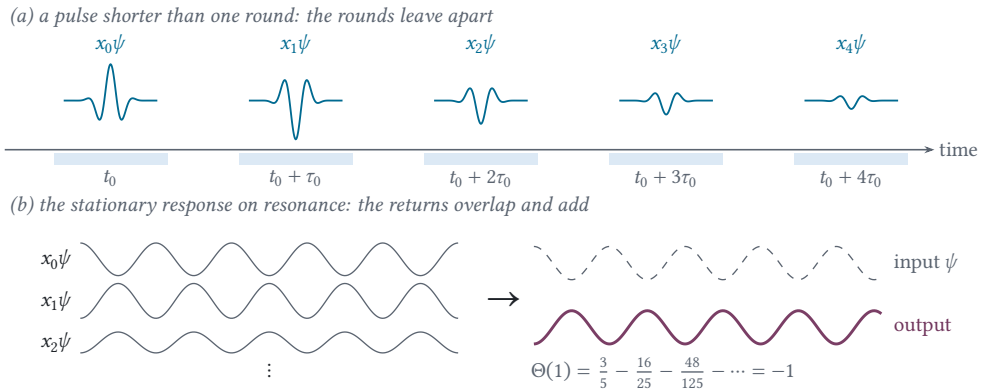


Fig. 42. A short pulse leaves in distinct time bins, with exit amplitudes $x_k u$. At one frequency the same exits contribute to one outgoing mode and add coherently.

Putting the bypass beside the ring gives the exits of `withReference`. Prepare its external qubit in $|+\rangle$, run the two arms, and apply H to compare them. At resonance $z = 1$, the single-frequency response is -1 and the difference port has probability $|1 - (-1)|^2/4 = 1$. For a short pulse the first time bin contributes $(1 - t)^2/4$ to this port, and the later bins contribute their probabilities separately:

$$\Pr(b = 1) = \frac{1}{4}\left((1-t)^2 + \sum_{k\geq 1} s^4 t^{2(k-1)}\right) = \frac{1}{4}\left((1-t)^2 + 1 - t^2\right) = \frac{1}{5}.$$

These are the coherent and measured predictions from Section 4. The scalar ring passes agreement, but the two-arm body fails: its immediate exit contains the bypass and its later exits do not. Forming that bypass uses a coherent sum of isos before channelization, an operation unavailable to a channel context.

The two experiments specify different input and output modes, one frequency or orthogonal time slots. A finite wave packet and observation window require their own specification before these predictions can be compared.

I Observing the response

For a fixed $|z| = 1$, the phase iso $P_z^c = \{s \leftrightarrow z \cdot s\}_{s \in \mathscr{B}(c)} : c \leftrightarrow c$ gives the marked source body $\beta;(I_b \oplus P_z^c)$. The bypass caller is

$$H;\mathsf{trace}_1\big(\texttt{withReference};(I_{\texttt{Qubit}} \oplus P_z^1)\big);H.$$

Thus the phase variation used in the body of the paper is an ordinary finite iso construction.

The controlled body. For $|z| = 1$ and $\delta \in \mathbb{R}$, let the reference space be $H_r = \mathbb{C}^2$, and put $P_\delta = \mathrm{diag}(1, e^{i\delta})$. Use distributivity to identify $H_r \otimes (X \oplus Y)$ with $(H_r \otimes X) \oplus (H_r \otimes Y)$. The body in Figure 11 is the finite unitary

$$W_{z,\delta} = \big(I_{H_r \otimes X'} \oplus (P_\delta \otimes zI_Y)\big)(I_{H_r} \otimes U).$$

It has external type $H_r \otimes X \to H_r \otimes X'$ and returning summand $H_r \otimes Y$. For reference basis value $r = 0$ its returning phase is z, and for $r = 1$ it is $ze^{i\delta}$. It never changes that basis value. Its kth exit on the two sectors is therefore respectively $z^k x_k$ and $(ze^{i\delta})^k x_k$. The closed gate is

$$|0\rangle\langle 0|_r \otimes \Theta(z) + |1\rangle\langle 1|_r \otimes \Theta(ze^{i\delta}).$$

For reference input $|+\rangle$ and unit data u, the closed gate returns

$$\frac{|0\rangle_r \otimes \Theta(z)u + |1\rangle_r \otimes \Theta(ze^{i\delta})u}{\sqrt{2}}.$$

Applying $S^\dagger$ and H gives data vectors on reference outcomes $0, 1$: $(\Theta(z)u - i\Theta(ze^{i\delta})u)/2$ and $(\Theta(z)u + i\Theta(ze^{i\delta})u)/2$. Their squared-norm difference is $\mathrm{Im}\langle\Theta(z)u, \Theta(ze^{i\delta})u\rangle$. Taylor expansion at $\delta = 0$ gives the displayed derivative readout in Figure 11.

Thus the straight reference line in the drawing denotes a tensor factor carried on both alternative sheets. Its controlled phase acts on each return. No coherent control of a channel hole is used.

At $z = 1$, this construction compares $\Theta(1)u$ and $\Theta(e^{i\delta})u$. For the scalar response $(3 - 5z)/(5 - 3z)$, choose $\delta = \pi/2$. The imaginary overlap is $8/17$, so the two outcome probabilities are $25/34$ and $9/34$. This finite-phase readout is the exact overlap, whereas the derivative formula is its first-order expansion at $\delta = 0$. This specializes Figure 11 to the controlled body $W_{1,\delta}$.